Systemadministration

Introduktion

Lagring

Agenda

1. Memory
2. Magnetband
3. Hårddisk
4. SSD
5. Gränssnitt
6. Protokoll
7. Chipset & Bandwidth
8. Cache

1. Memory

Memory Types

Flera olika typer för att lagra information

Volatile

- förlorar sitt innehåll när strömmen bryts

Non-volatile

- ’permanent’ lagring

Memory Types

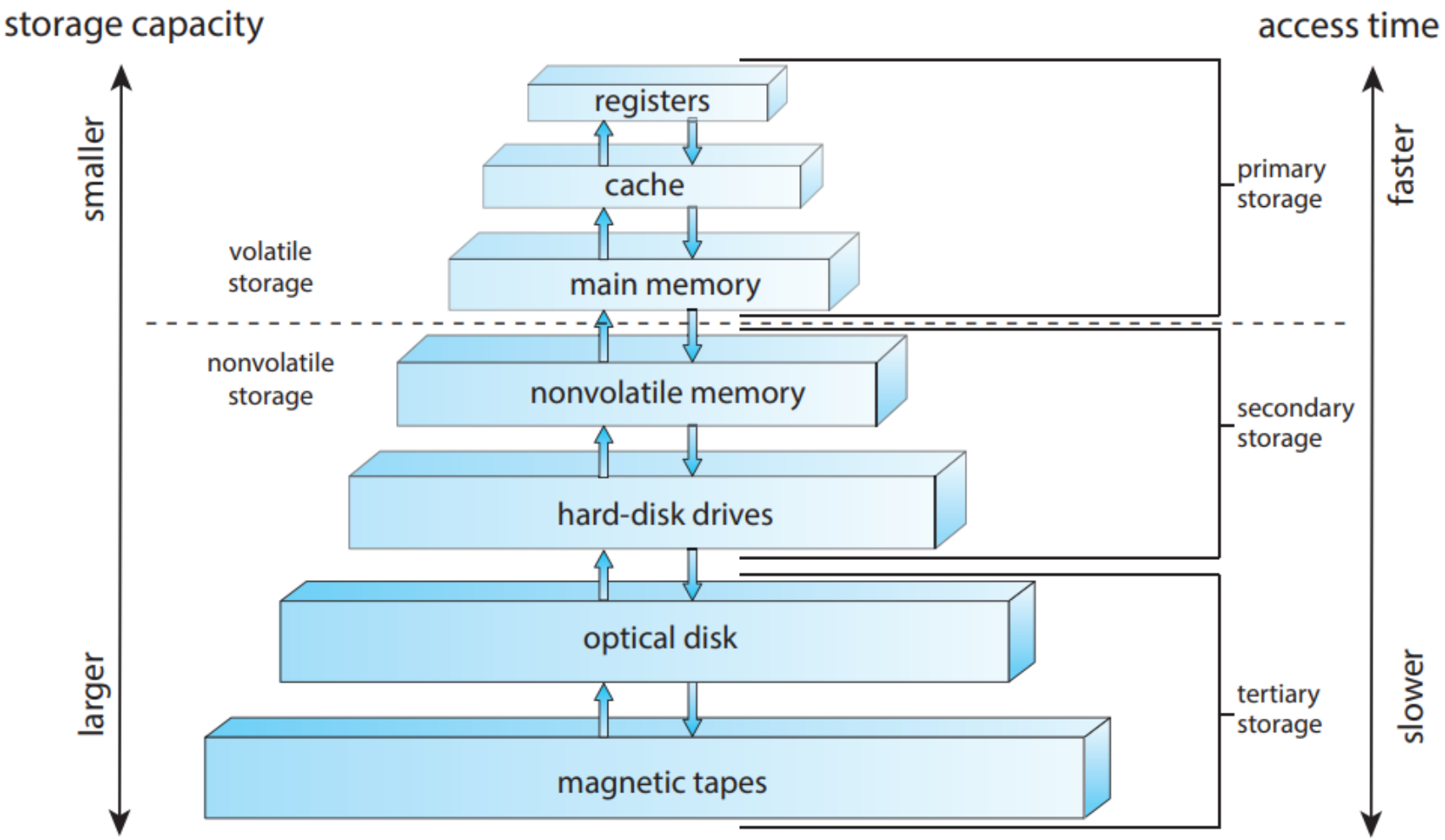
Random access

- Lika snabbt oavsett vart data finns

Block-based

- Data lagras i en samling av bitar/bytes
- Läser/skriver som minst ett block i taget

Memory Hierarchi



[1]

2. Magnetic Tape

Magnetic Tape

Magnetband

Tidigt sekundärminne

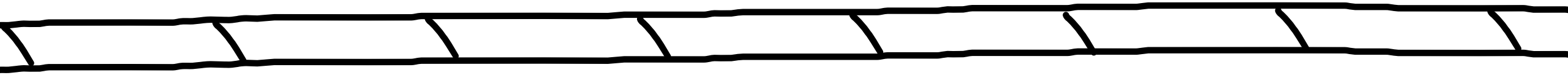
Bank of IBM 729 magnetic tape drives at the Computer History Museum IBM 1401 exhibit, by ArnoldReinhold, 2016, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:IBM_729_tape_drives.agr.jpg). CC BY-SA 3.0.

Magnetic Tape

Magnetband

Data hanteras i **block**, en samling bytes

Läser/skriver ett block i taget



3. Hard Disk Drive

Hard **D**isk **D**rive
Hårddisk

Modern **HDD**

- 1-20 TB
- 50-250* MB/s

Mycket stöt & vibrations känsliga

A pair of Seagate Barracuda hard drives, by Ashley Pomeroy, 2020, Wikimedia (https://commons.wikimedia.org/w/index.php?curid=93199385). CC BY-SA 4.0.

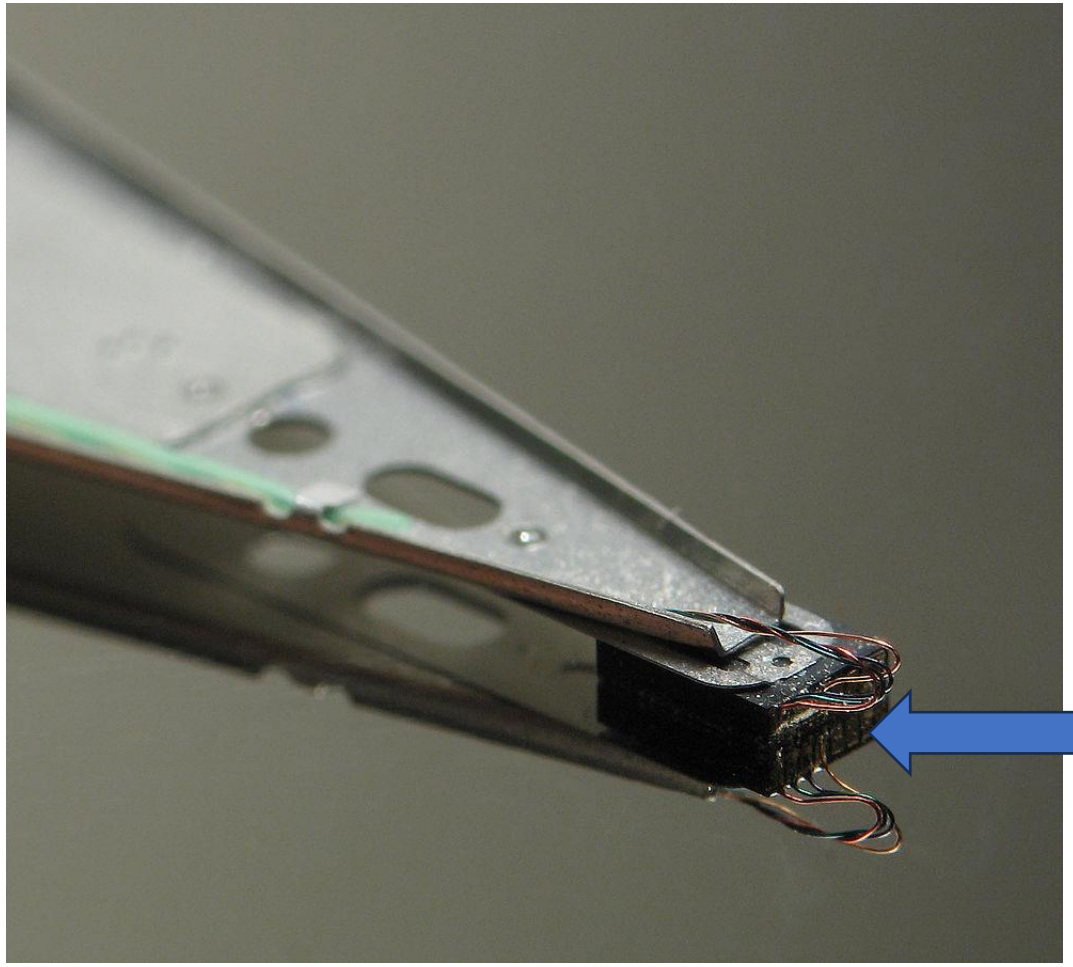
*i bästa fall

Hard Disk Drive
Hårddisk

Platters

Innenansicht einer Festplatte. Hier mit 512 MB, was aber für den Grundaufbau nicht so relevant sein dürfte, by SPBer, 2007, Wikimedia (https://en.wikipedia.org/wiki/File:Innansicht_Festplatte_512_MB_von_Quantum.jpg). CC BY-SA 3.0.

Hard Disk Drive
Hårddisk

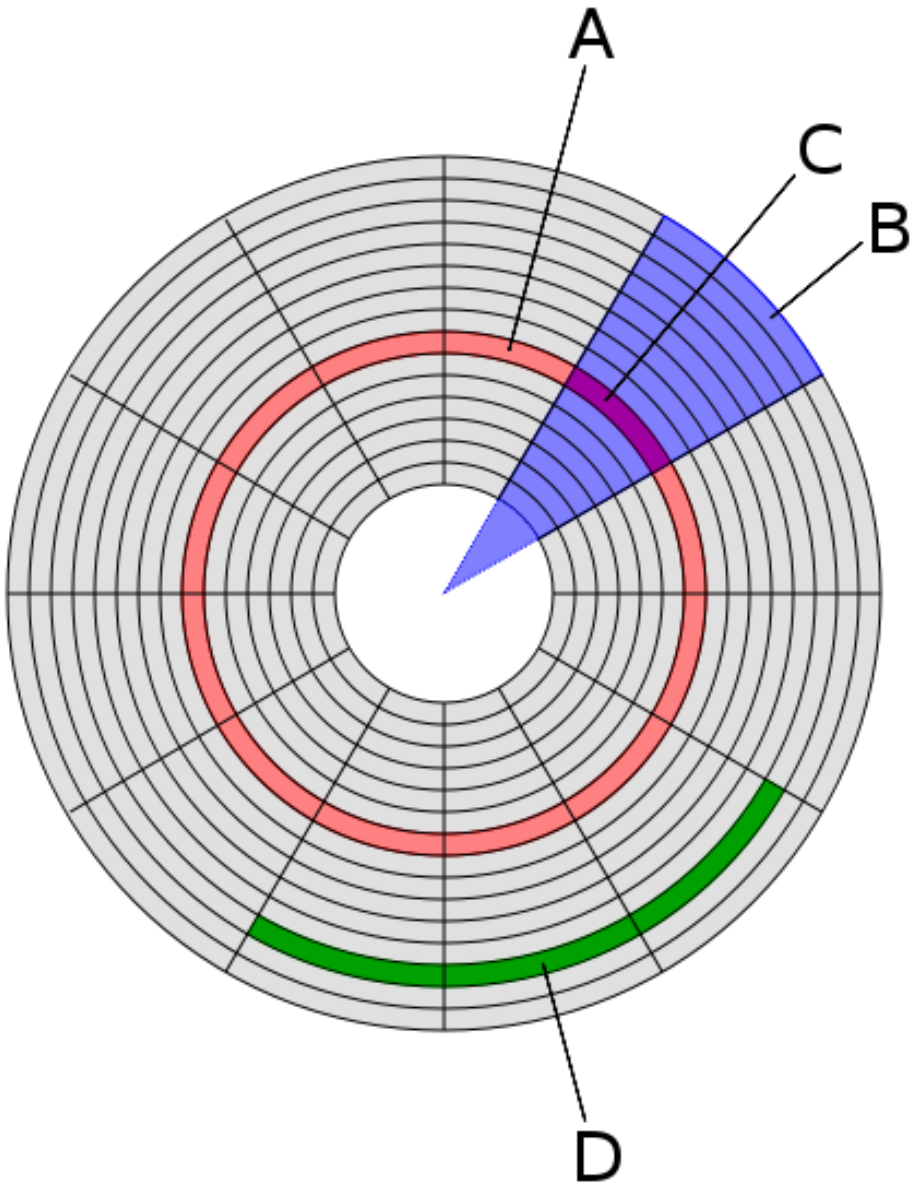


R/W-head

Hard drive head, by Andrew Magill, 2006, Flickr (https://www.flickr.com/photos/amagill/89623319/). CC BY 2.0.

Hard Drive Geometry

A) **Track**
C) **Sector**
D) **Filesystem cluster**

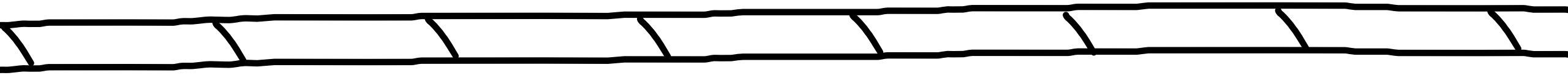


Disk structure showing a track (A), a geometrical sector (B), a track sector (C) and a cluster of sectors (D). The term "sector" in the context of computer engineering usually refers to (C), by Heron2/MistWiz, 2008, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:Disk-structure2.svg). Public Domain.

Sectors

En HDD presenterar sin lagring som en lång serie av **sectorer**

Äldre diskar: **512 Bytes**
Nyare diskar: **4096 Bytes**



Head Crash

När läs-/skrivhuvudet kolliderar

→ Permanent skada

Head crash in a new model hard disk, by Alchemist-hp, 2009, Wikimedia Commons (https://en.wikipedia.org/wiki/File:Hard_disk_head_crash.jpg). CC BY-SA 3.0.

3M Science. Applied to Life.™

February 2016

Clean Extinguishing Agent System Noise and Hard Disk Drive (HDD) Failure FAQs

What we know

From 2008 through 2013 there have been multiple documented events per year where actuation of an inert gas clean agent fire suppression system caused subsequent failure of hard disk drives (HDDs).[1] There are no reported instances of a halocarbon clean agent system discharge, including systems with 3M™ Novec™ 1230 Fire Protection Fluid, that have resulted in HDD damage.

What causes HDD failure during a discharge?

The cause of HDD failure is acoustic noise at a specific frequency and decibel (dB) level impinging on the flexible stainless steel cover of an HDD case. Such impingement transmits vibration into the spindle and actuator axis, causing the read/write element to misalign and fail. Current generation HDD cannot tolerate more than 12 nanometers (0.00000045 in.) of misalignment.[2] Room pressure and temperature changes resulting from a CEA discharge do not affect HDD performance or failure rates.

BBC

Fire drill knocks ING bank's data centre offline

12 September 2016

ING's Romanian customers were unable to access their money for much of Saturday

The Register

Noise from blast of gas destroys Digiplex data depot disk drives

Nordic Nasdaq knocked as deleterious decibels crashed servers

Chris Mellor
Thu 26 Apr 2018 // 13:57 UTC

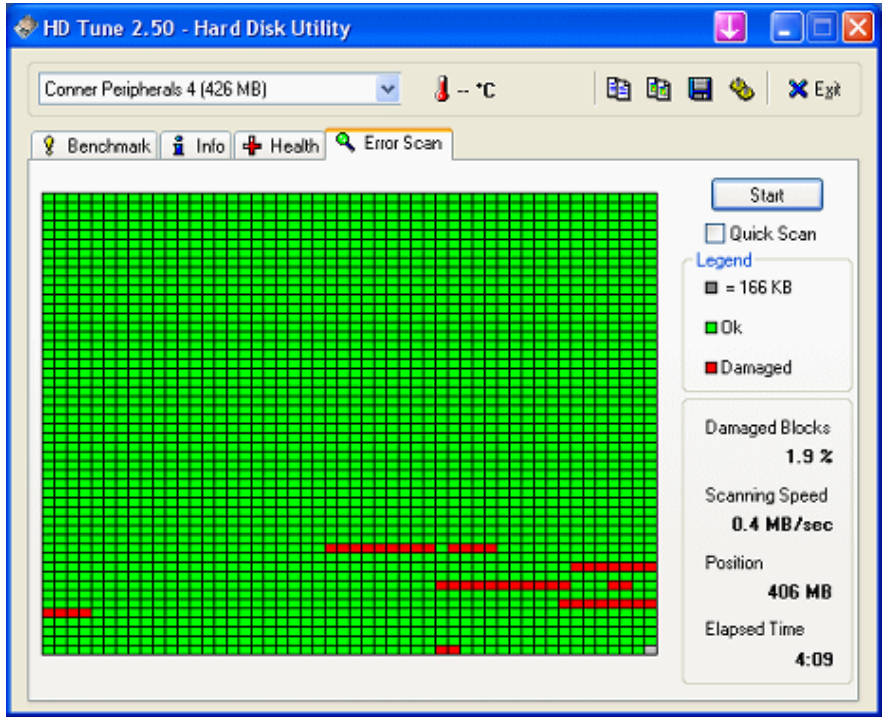
It's a bit rich when a fire prevention system designed to protect a data centre kills its servers, but that's what happened when inert gas was injected into a Nasdaq Digiplex bit barn in Sweden.

The gas release – on 18 April – caused a loud, high-pitched whistle, which resulted in vibrations destroying server disk drives, the servers crashing and the Nasdaq Nordic exchange going offline until 2pm local time. A backup system had to be brought into action by Nasdaq, which rents space at the Digiplex facility in Väsby, near Stockholm, with its own kit installed there.

Bad Sector

Sektor som inte går att läsa tillbaka pga.

- Fysisk skada
- Mjukvarufel



Run the Error Scan, by HD Tune, n.d., HD Tune (http://www.hdtune.com/faq_1.html). Fair use/I enlighet med citaträtt

Shattered Hard Disk Platters, by techmsg, 2012, Flickr (https://flickr.com/photos/techmsg/7875979862/). CC BY 2.0.

Disk Parking

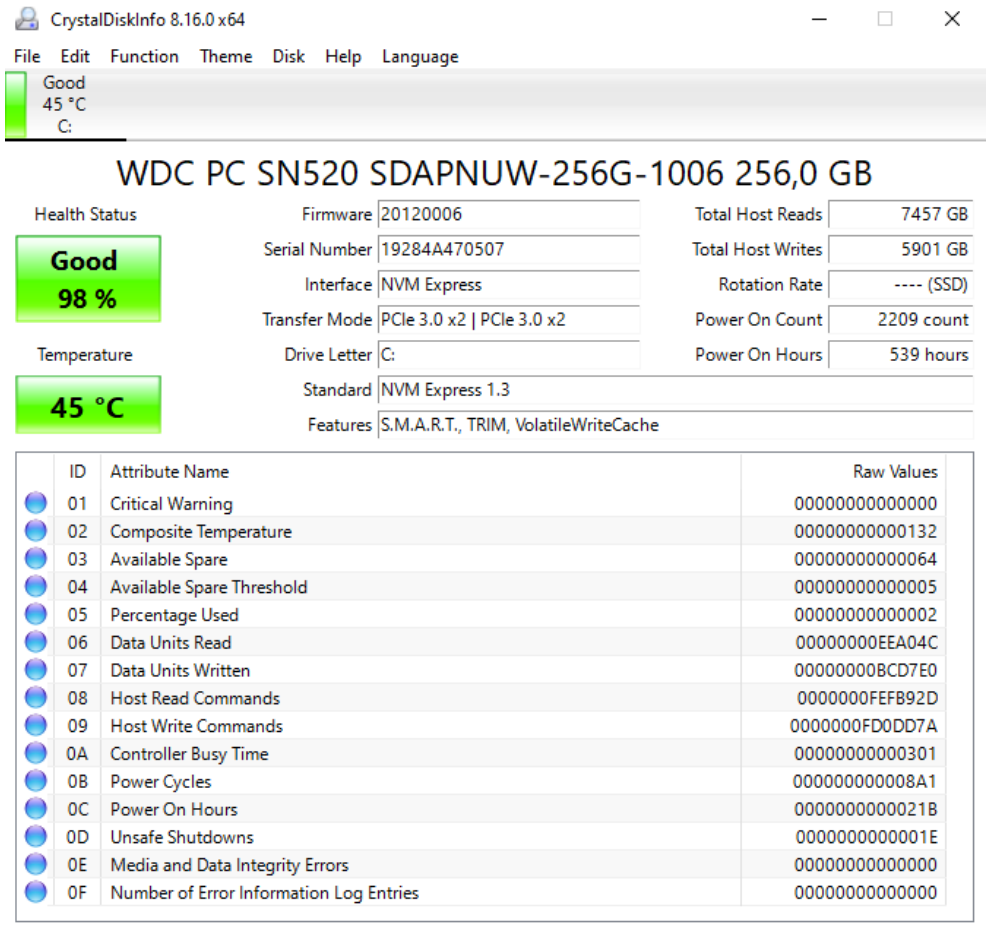
För att skydda mot stötar när disken är avstängd ’**parkeras**’ den

A 2.5" hard drive that has been opened, exposing its inner workings. This is a 500GB Western Digital Scorpio Blue hard drive with SATA connections. This 2.5" hard drive is common in laptops, by Evan-Amos, 2013, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:Laptop-hard-drive-exposed.jpg). CC BY-SA 3.0.

Self-**M**onitoring, **A**nalysis and **R**eporting **T**echnology

S.M.A.R.T. övervakar hårddiskar och SSDer

- Loggar fel
- kan varna för förutspått haveri



4. Solid-State Drive

Solid-**S**tate **D**rive

SSD

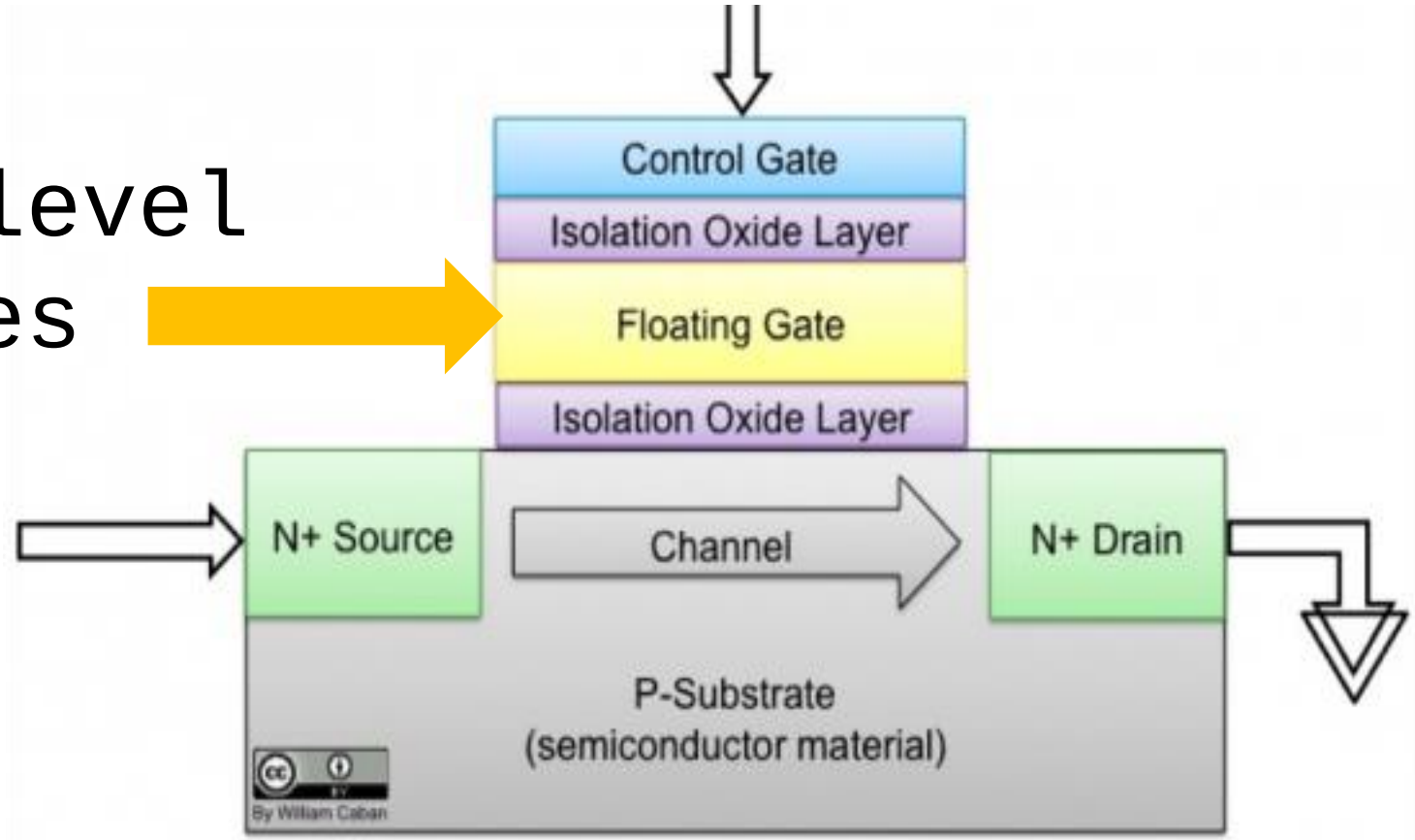
- 250 GB - 8 TB
- 500-10 000* MB/s

*i bästa fall

Flash Cell

Read stored charge

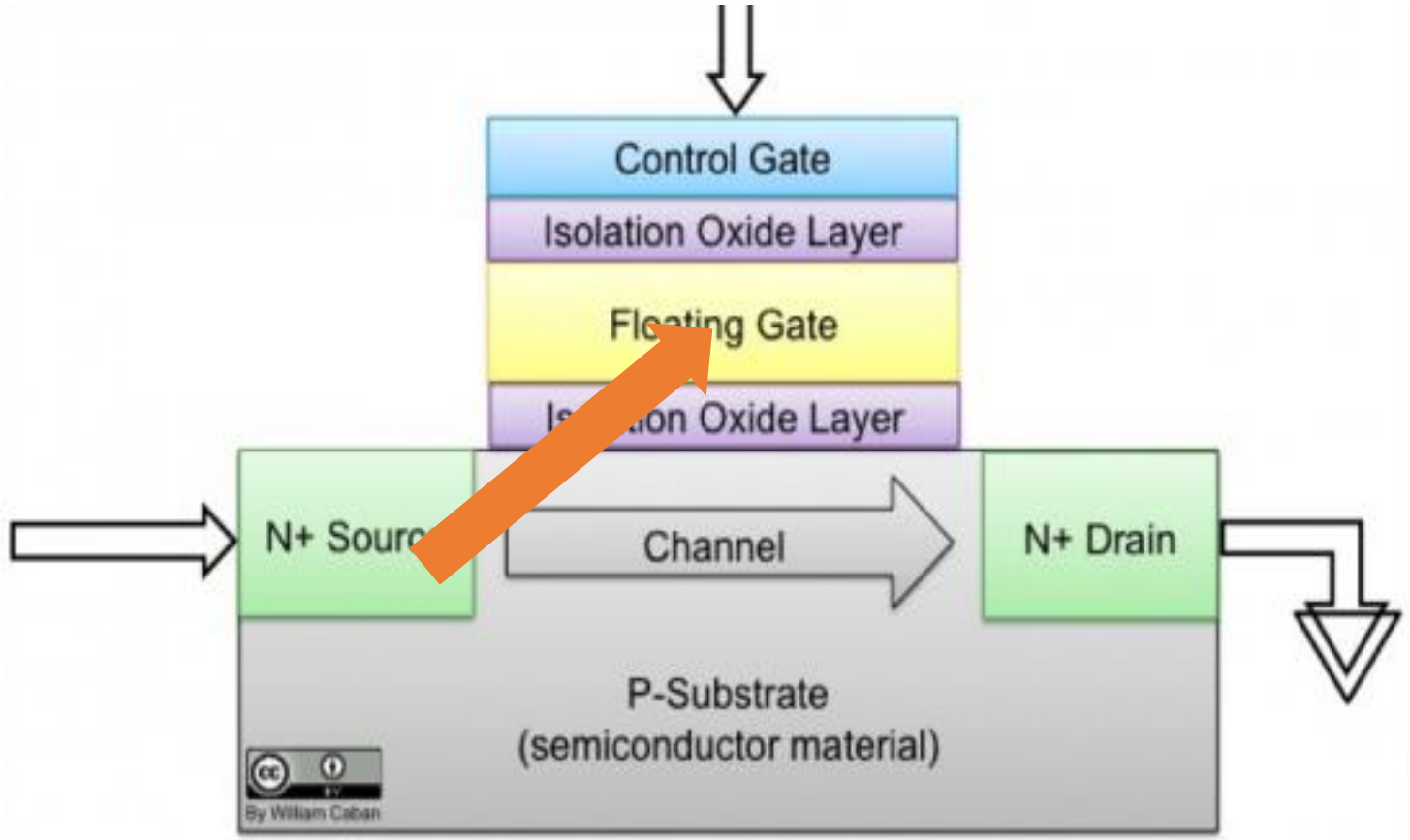
Charge level indicates 1 or 0



by William Cabon-Babilonia, 2014, Cisco Blogs (https://blogs.cisco.com/datacenter/decoding-ucs-invicta-part-2). CC BY.

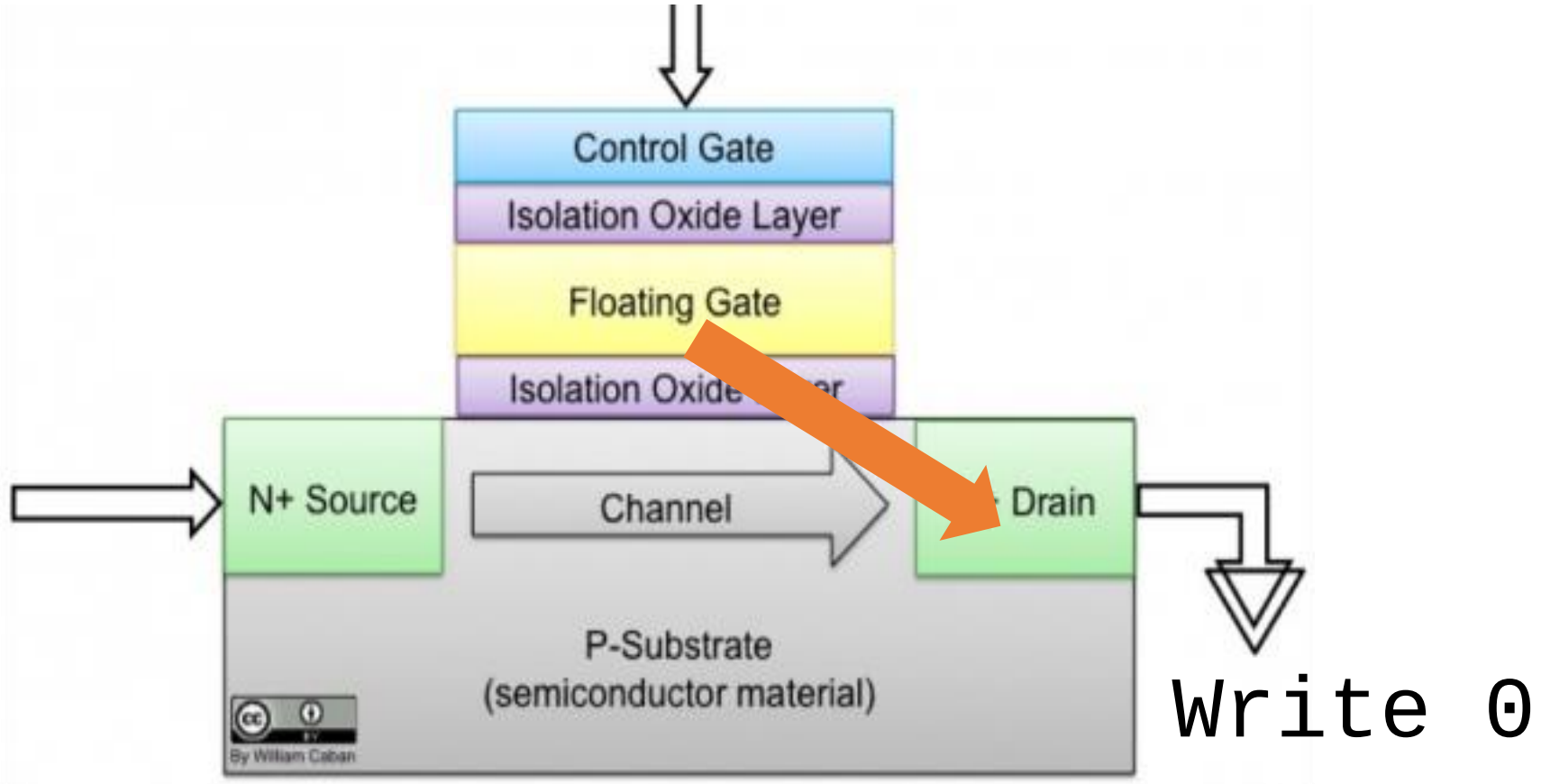
Flash Cell

Write 1



by William Cabon-Babilonia, 2014, Cisco Blogs
(https://blogs.cisco.com/datacenter/decoding-ucs-invicta-part-2). CC BY.

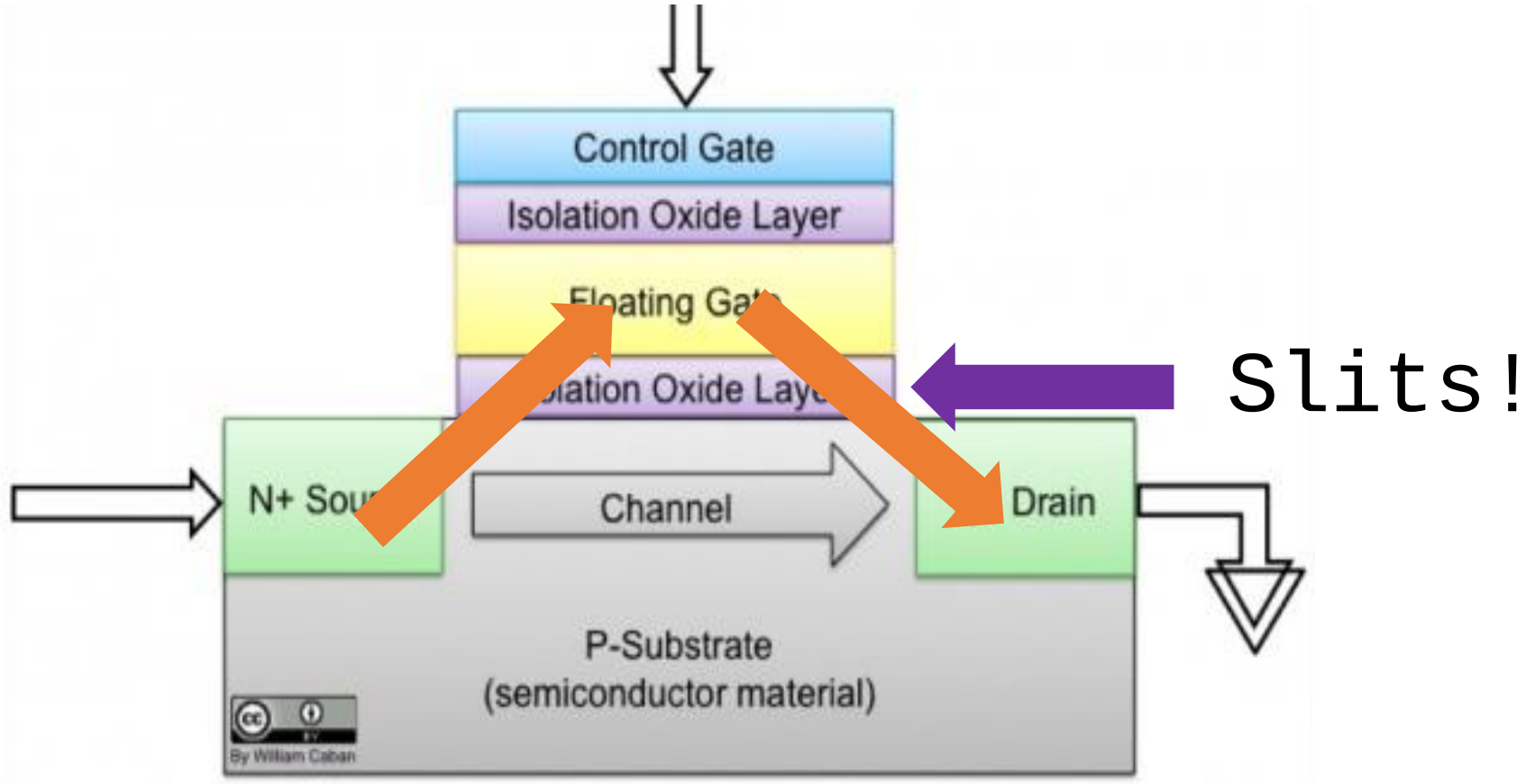
Flash Cell



Write 0

by William Cabon-Babilonia, 2014, Cisco Blogs
(https://blogs.cisco.com/datacenter/decoding-ucs-invicta-part-2). CC BY.

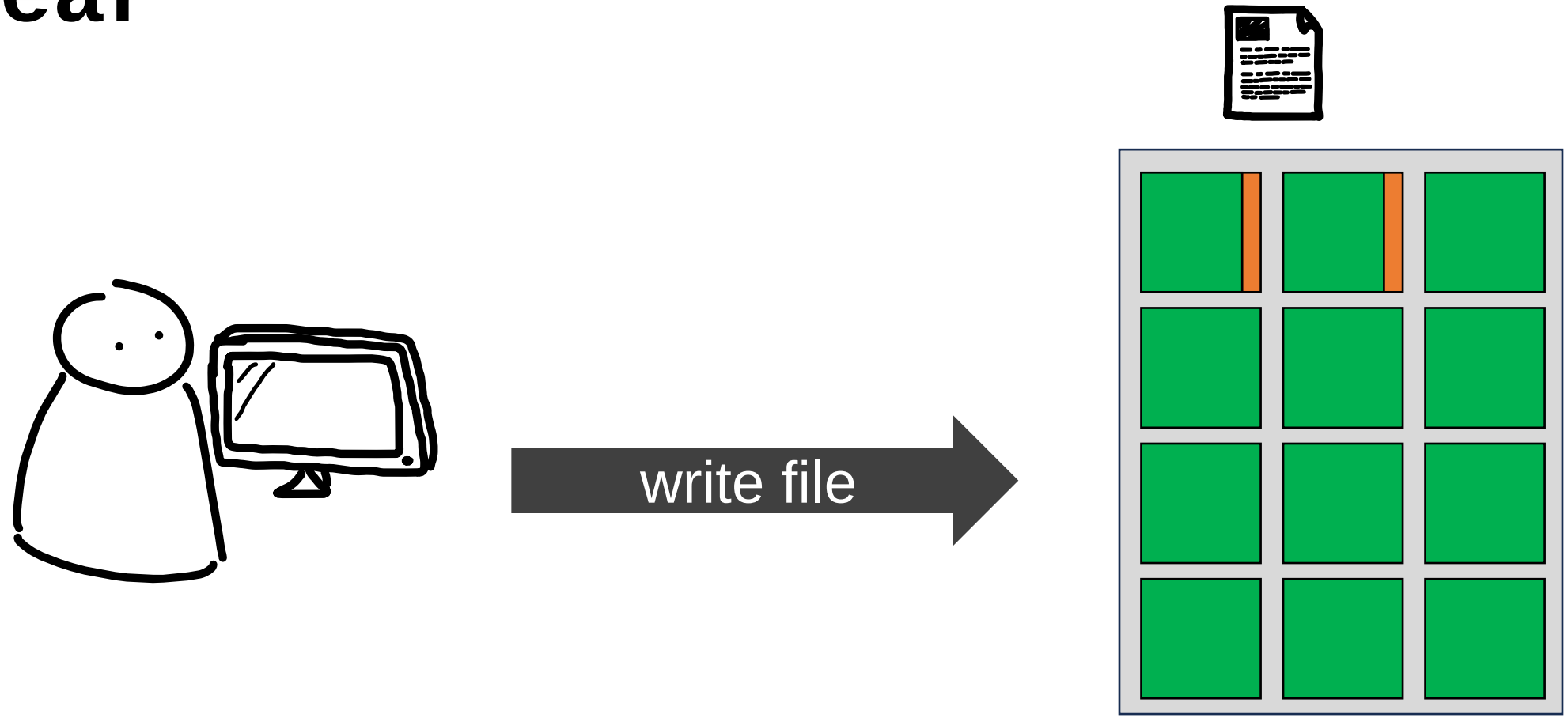
Flash Wear



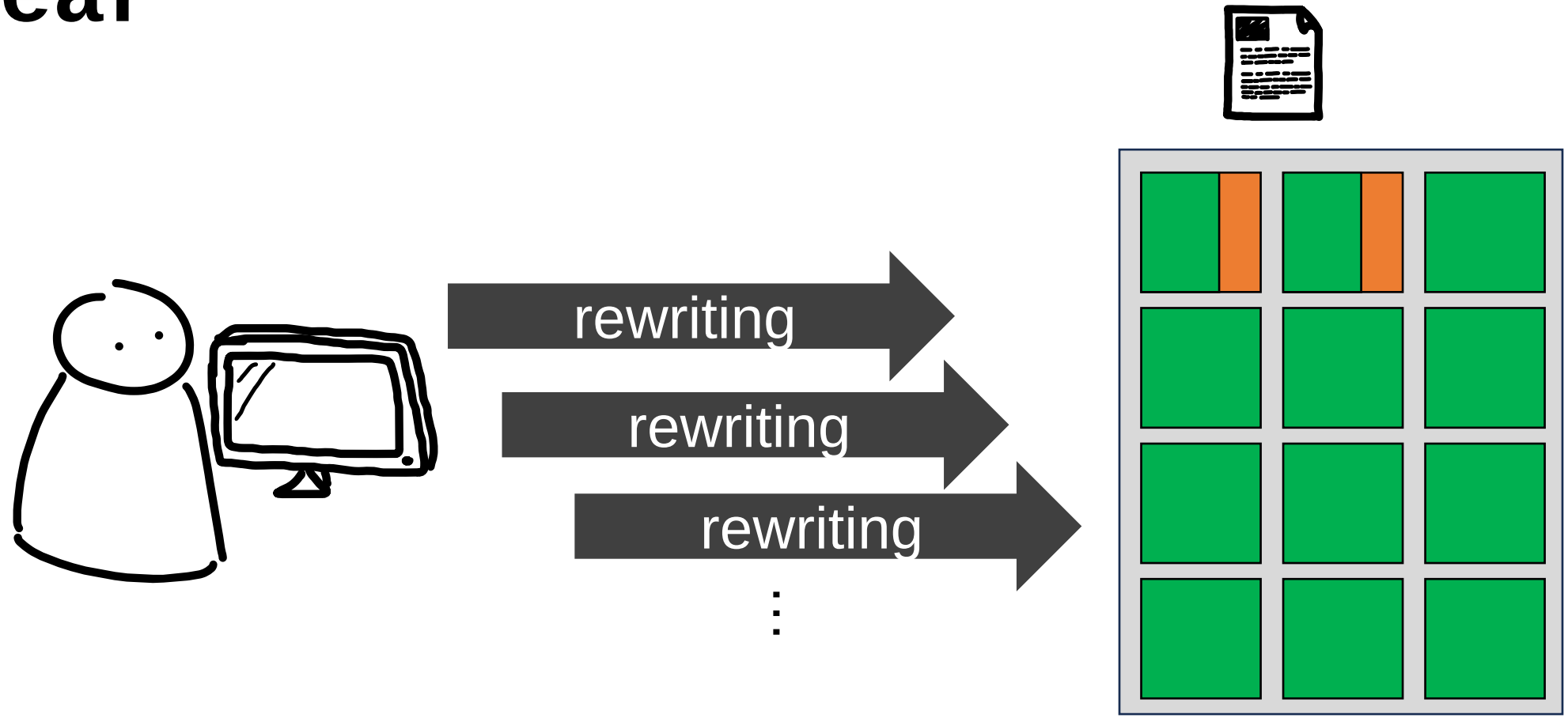
by William Cabon-Babilonia, 2014, Cisco Blogs
(https://blogs.cisco.com/datacenter/decoding-ucs-invicta-part-2). CC BY.

Wear

Wear

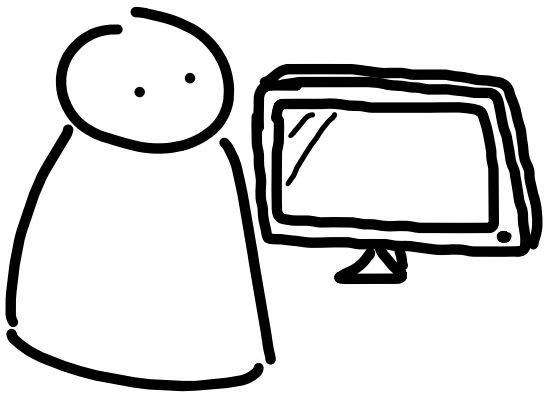


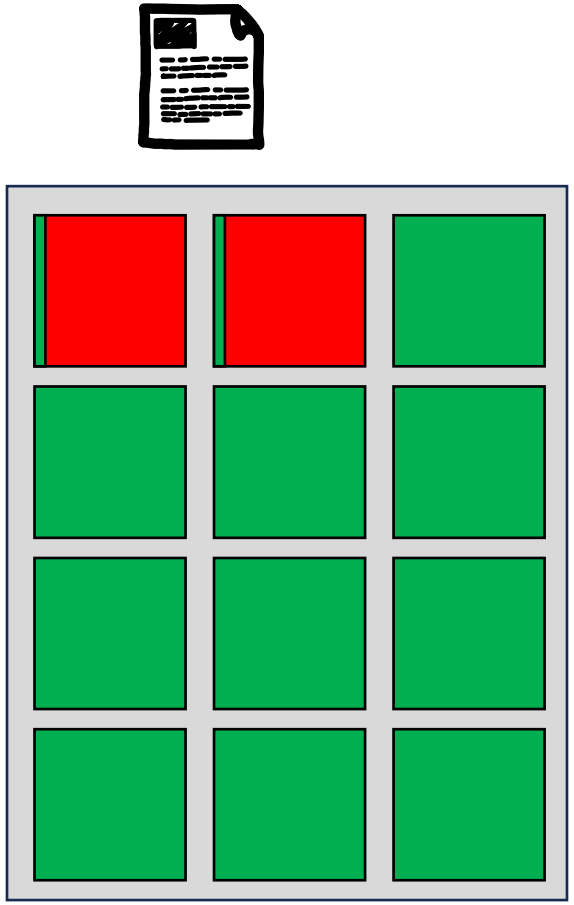
Wear



12 cell chip

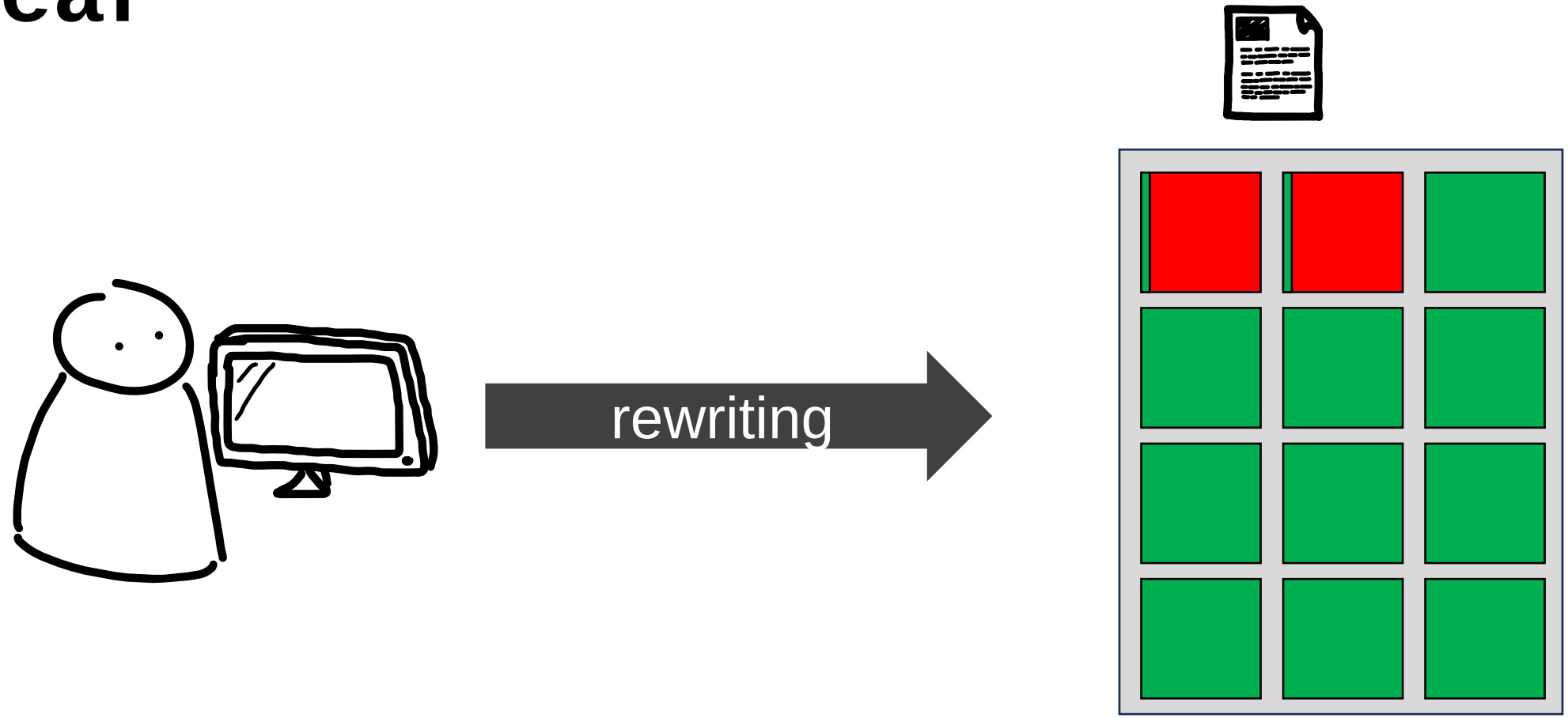
Wear





12 cell chip

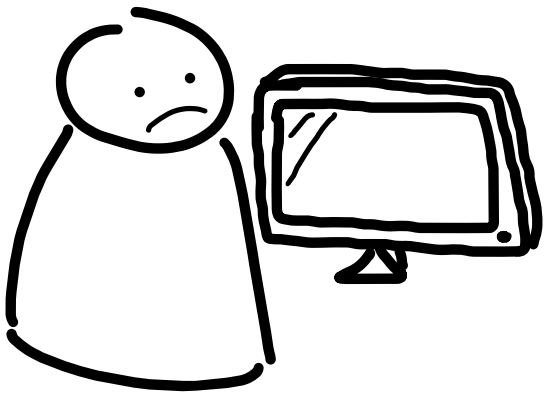
Wear

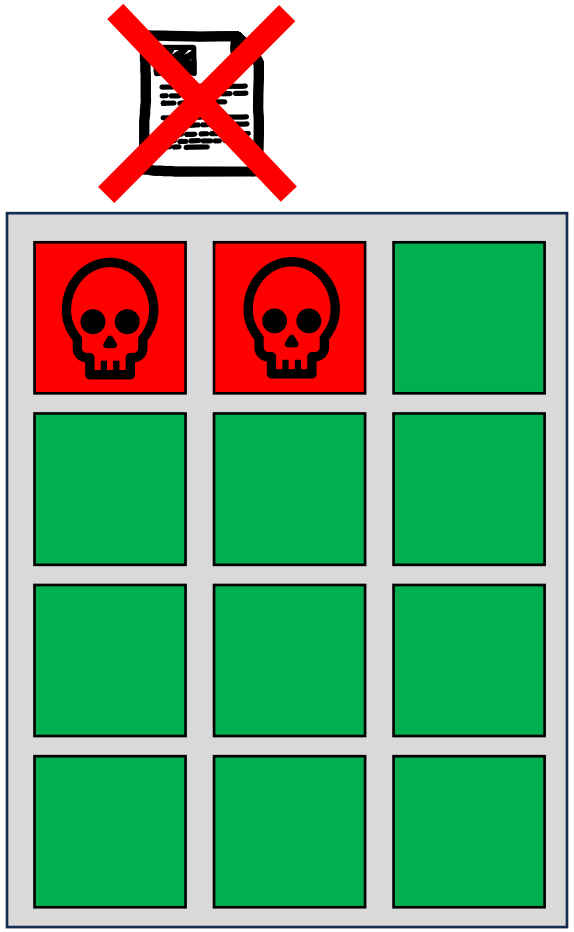


Wear

Cells no longer hold charge

File lost



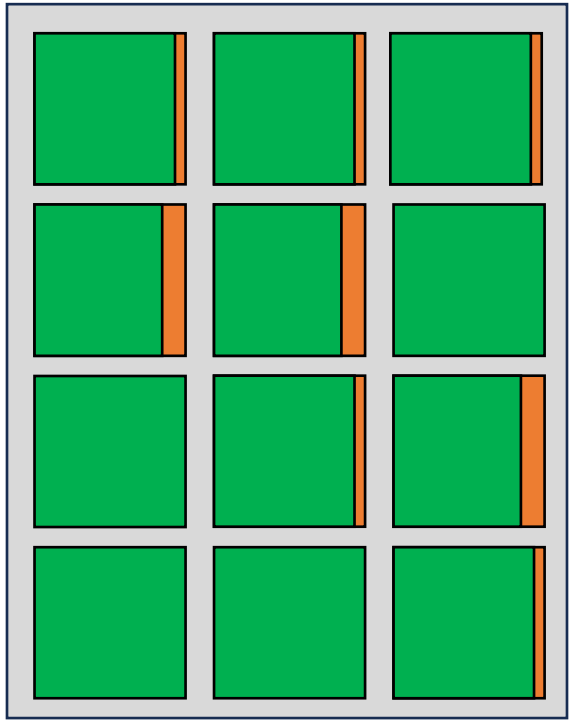


12 cell chip

Wear Leveling

SSDer sprider slitage jämnt

- Mappar dynamiskt om celler
- Flyttar data mellan celler

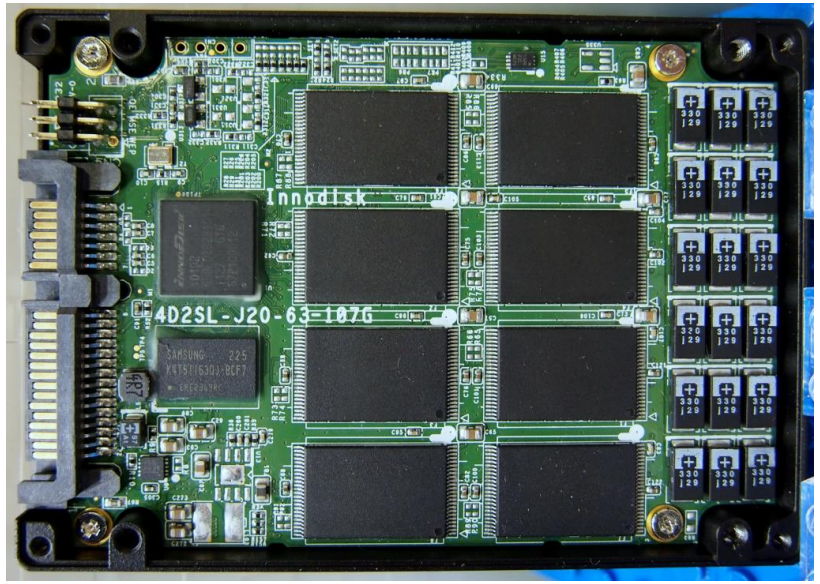


12 cell chip

Over Provisioning

Fler celler än vad som visas för datorn

Används för att förlänga livslängden av **SSD**n



Embedded World 2014: Innenleben eines Innodisk-Industrial SSD-Laufwerk, by Ordercracy (Thomas Springer), 2014, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:Embedded_World_2014_SSD.jpg). CC0 1.0.

Cell Endurance

Kan skriva om hela lagringsytan 10 gånger/dag i 5 år

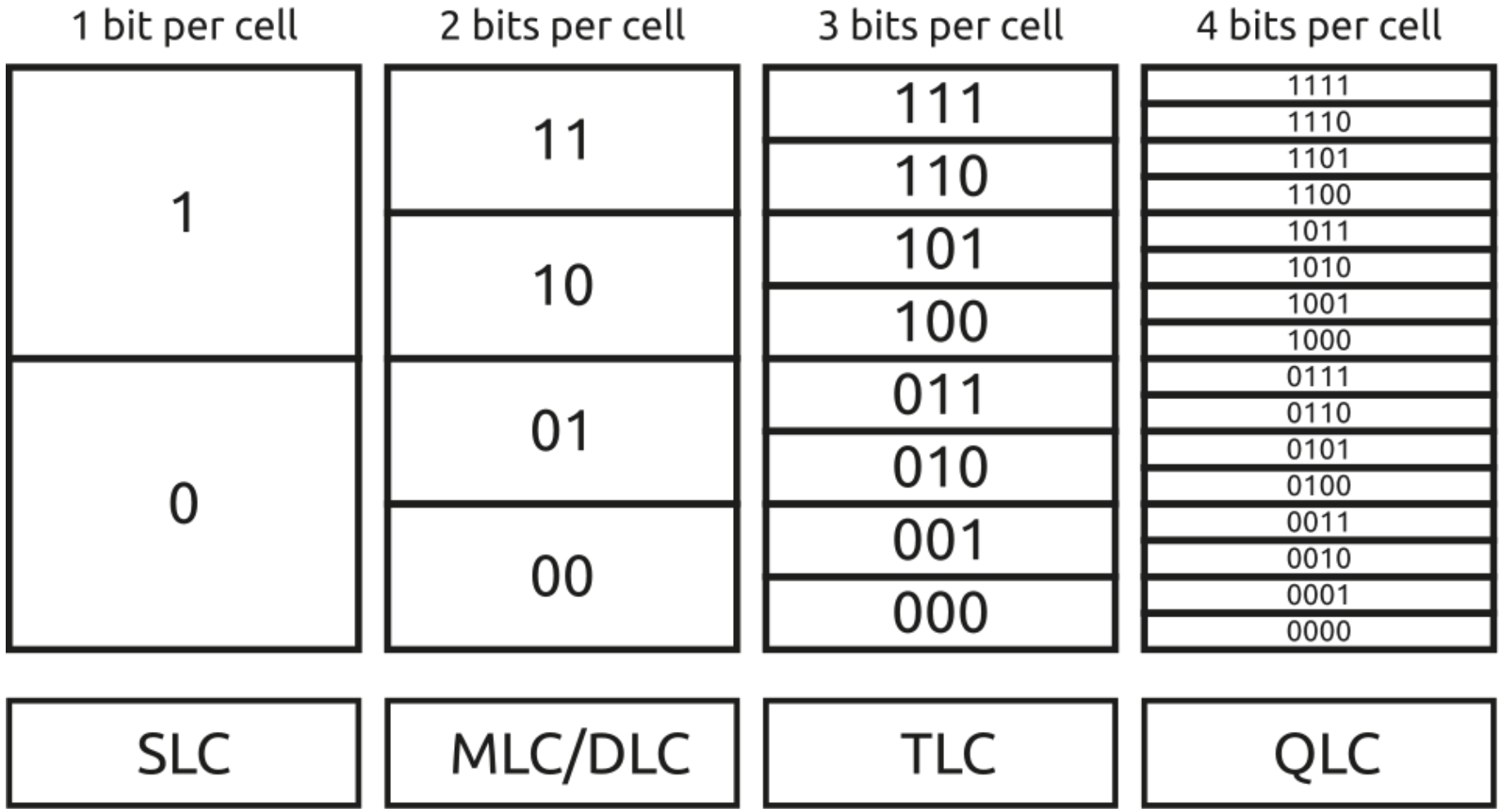
Billiga konsument diskar klarar betydligt mindre

Nytro 3050 Series

Capacity	Up to 15.36 TB
Endurance	Up to 10 DWPD
Sequential Read	Up to 2,200 MB/s
Warranty	5 years

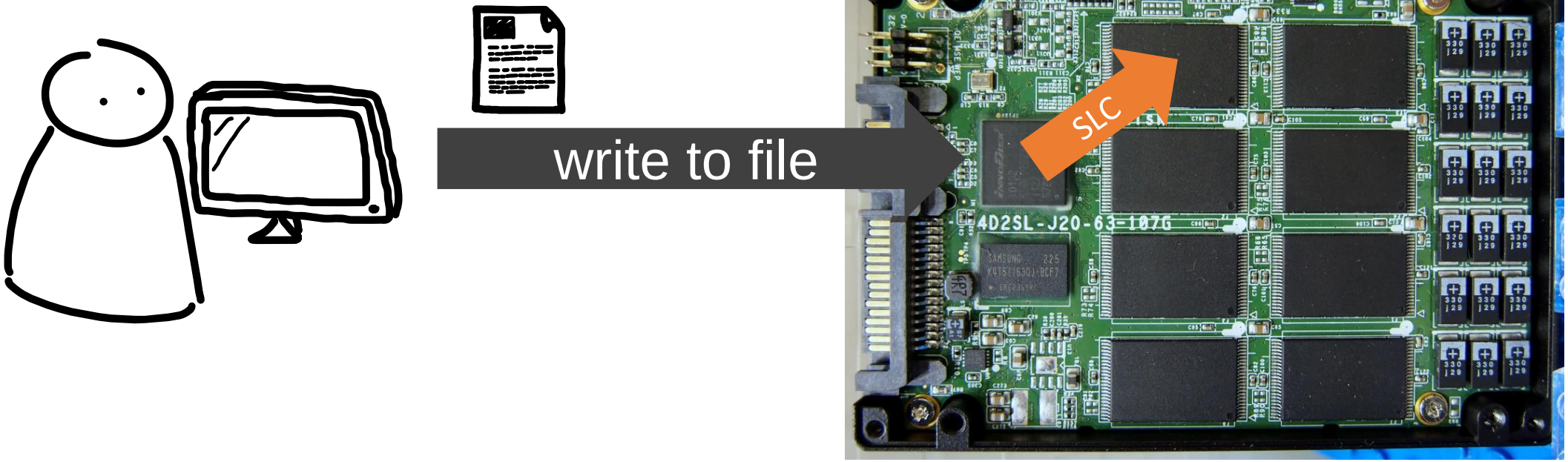
Nytro SAS SSD Series, by Seagate, u.å., Seagate (https://www.seagate.com/gb/en/products/enterprise-drives/nytro-sas-ssd/). Fair use/ I enlighet med citaträtt.

Multi-Level Cell



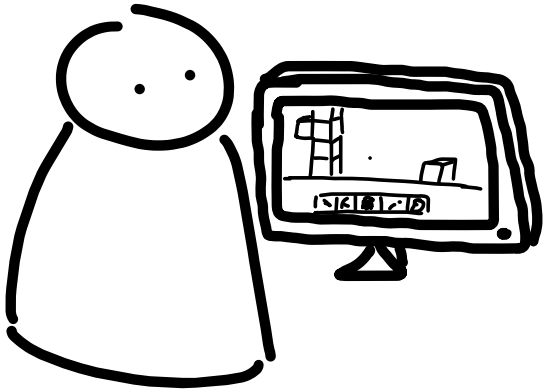
The cell types in flash memory SLC to PLC in comparison, by PantheraLeo1359531, 2021, Commons Wikimedia (https://commons.wikimedia.org/wiki/File:Cell_types_SLC-PLC_in_comparison_20211102.svg). CC0 1.0.

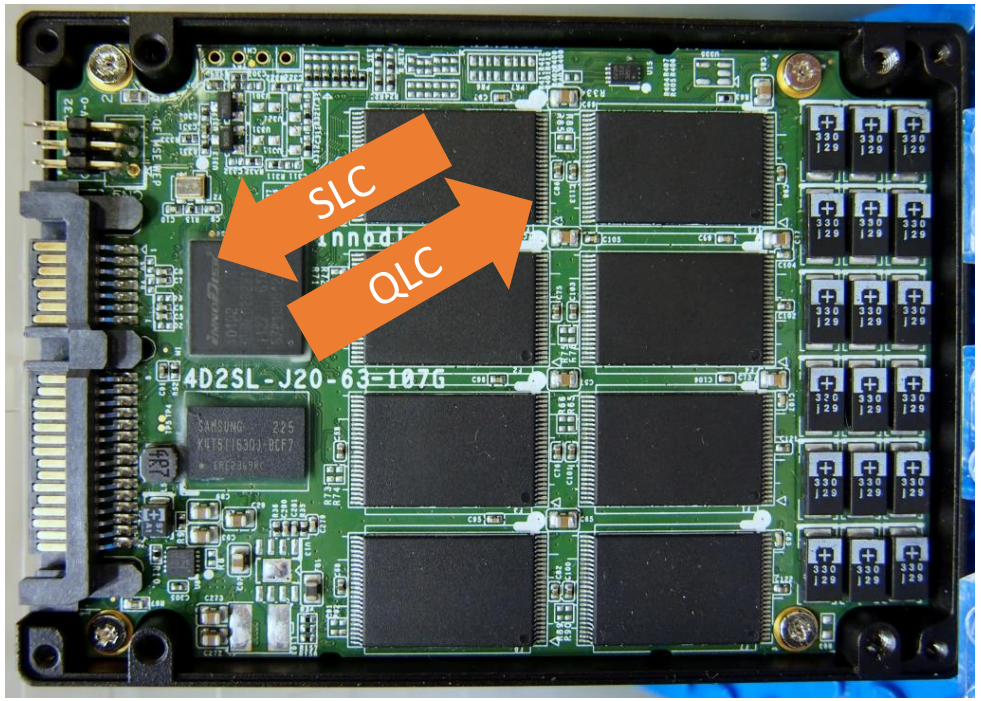
Multi-Level Cell



Embedded World 2014: Innenleben eines Innodisk-Industrial SSD-Laufwerk, by Ordercracy (Thomas Springer), 2014, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:Embedded_World_2014_SSD.jpg). CC0 1.0.

Multi-Level Cell





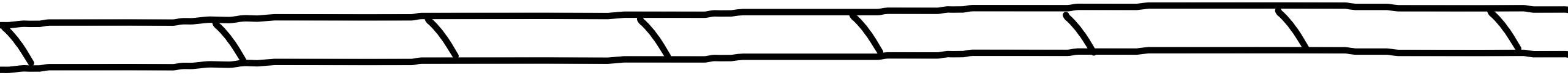
Embedded World 2014: Innenleben eines Innodisk-Industrial SSD-Laufwerk, by Ordercracy (Thomas Springer), 2014, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:Embedded_World_2014_SSD.jpg). CC0 1.0.

När **SSD**n inte används skrivs celler om till högre densitet

Sectors

En SSD presenterar sin lagring som en lång serie av **sectorer**

En **sector** = **4096 Bytes**



5. Gränssnitt

Protokoll & gränssnitt

Protocols & Connectors

Sekundärminne ansluts med

- Fysiska anslutning
- Protokoll

Ex.

Protokoll	Gränssnitt
SATA	SATA

SCSI (legacy)

Lanserades 1986

Snabb, dyr men tillförlitlig

by Ian Wilson, 2006, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:Centronics_50_SCSI_connector.JPG). CC BY-SA 3.0.

Parallel **ATA** (legacy)

Kallas även **IDE**

Ett billigare & långsammare alternativ till **SCSI**

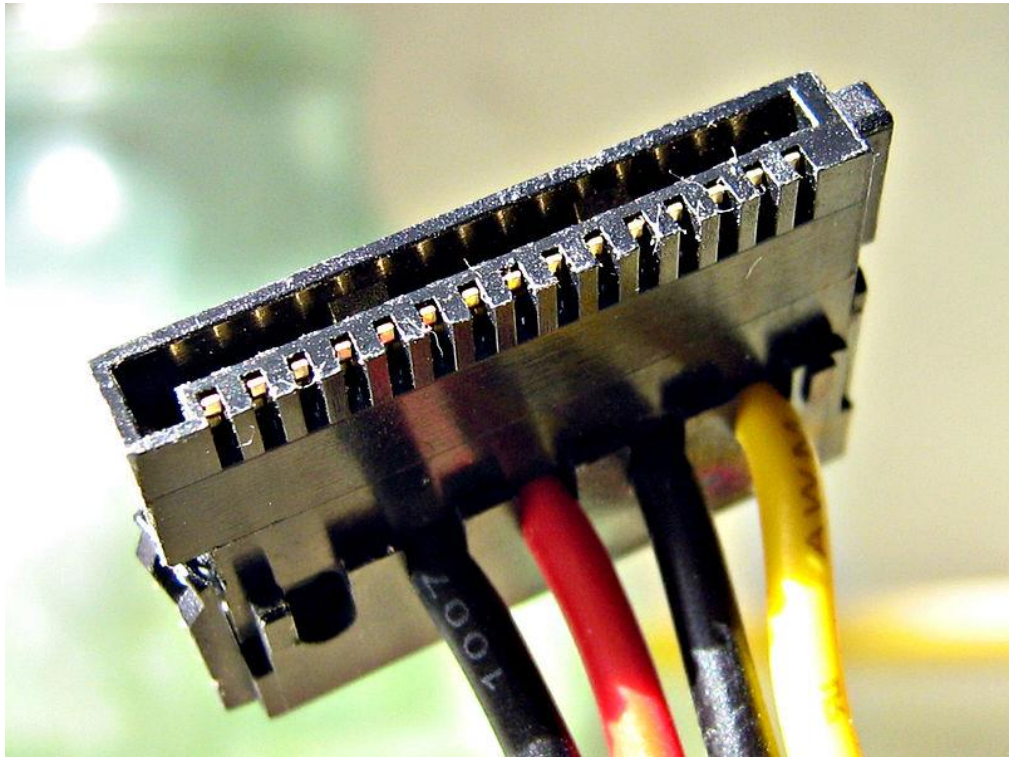
a parallel-ata cable for connecting devices to the motherboard of a computer, by Evan-Amos, 2010, Wikimedia Commons (https://en.wikipedia.org/wiki/File:PATA-cable.jpg). Public Domain.

Serial AT Attachment

SATA DATA

Разъём SATA, by T4bits~commonswiki, 2005, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:SATA_Data_Cable.jpg). CC BY-SA 3.0.

SATA POWER

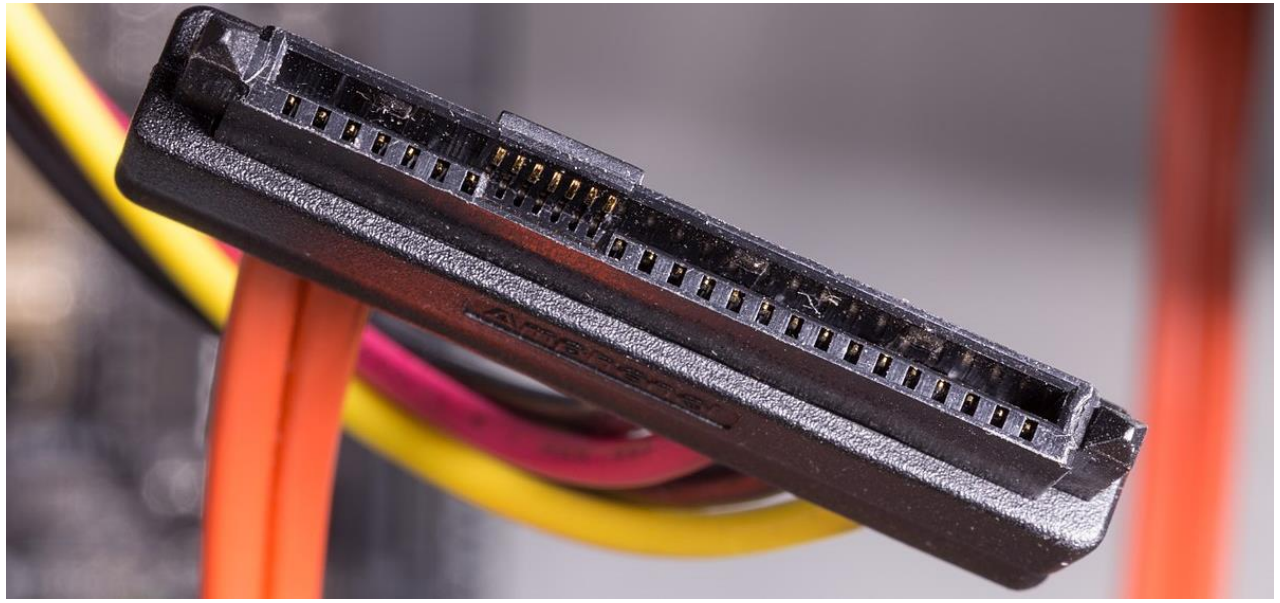


*A SATA power cable with 15-pin connector. * Date: 29th September 2004 23:05 * Camera: FUJIFILM FinePix S7000 * Exposure: 1/70 sec. * Focal Length: 8mm * Aperture: F/2.8 ISO: 400 * Photograph: ed g2s*, by Ed g2s, 2011, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:SATA_power_cable.jpg). CC BY-SA 3.0.

Serial Attached SCSI

SFF-8639 aka U.2 connector (male) to Mini-SAS HD adapter. PCI-4339B-1M-N2, by Kevinf28, 2016, Wikimedia Commons (https://en.wikipedia.org/wiki/File:SFF-8639.jpg). CC BY-SA 4.0 International.

SFF-8087 (internal mini-SAS) connector (4x SAS), by Dimitry Nosachev, 2016, Wikimedia Commons (https://en.wikipedia.org/wiki/File:SFF-8087_SMC.jpg). CC BY-SA 4.0 International.



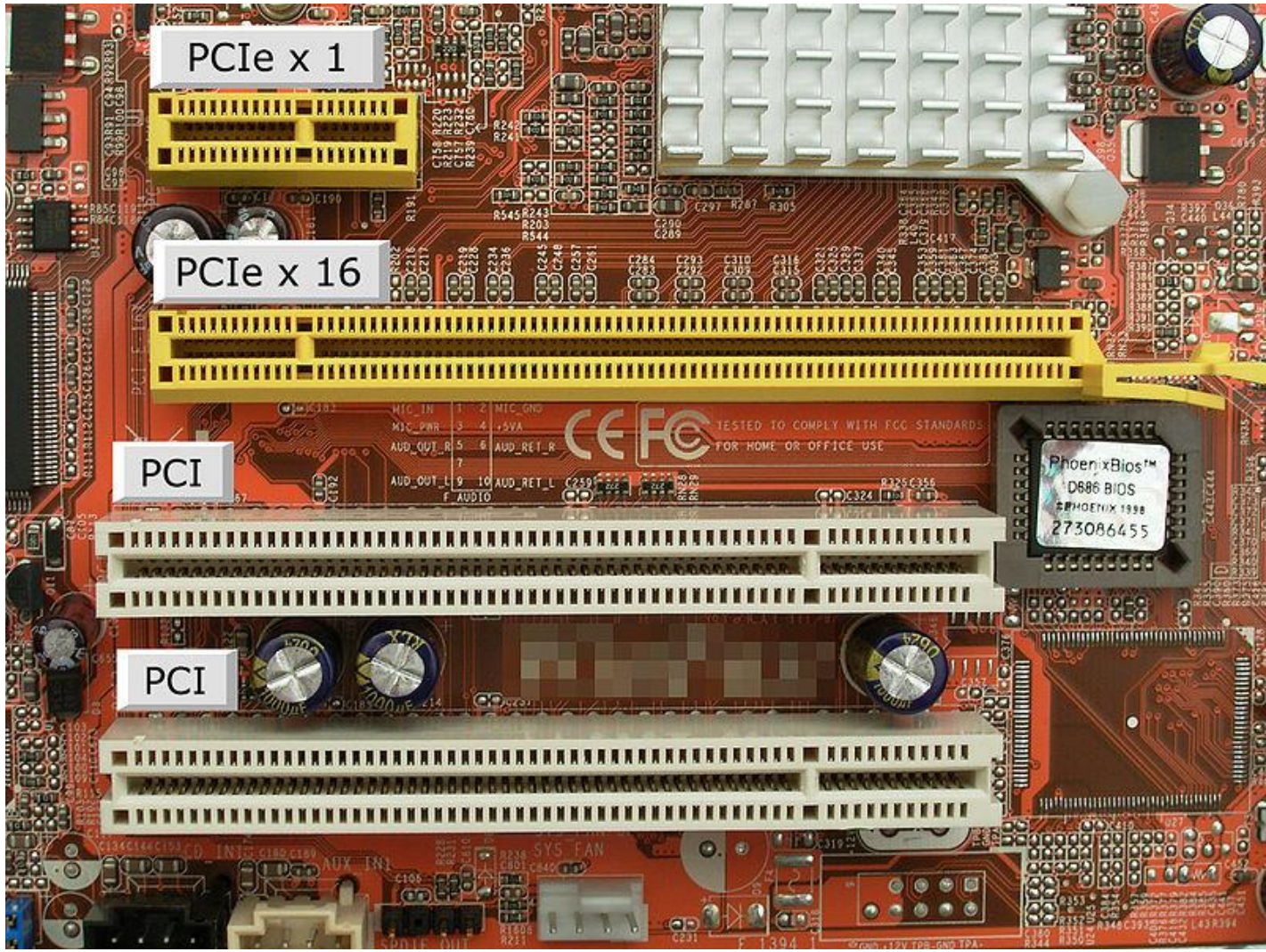
SFF-8482 SAS Plug with 7 pin SATA and 4-pin 8981 peripheral power connector on the other end. And Bokeh at f/22 as noticed later, by Tobias "ToMar" Maier, 2017, Wikimedia Commons (https://en.wikipedia.org/wiki/File:SFF-8482_SAS_Plug_-_Bokeh_at_f22.jpg). CC BY-SA 3.0.

PCI express

Ansluter insticksskort

Bandbredd beror på

- Lanes
- Version



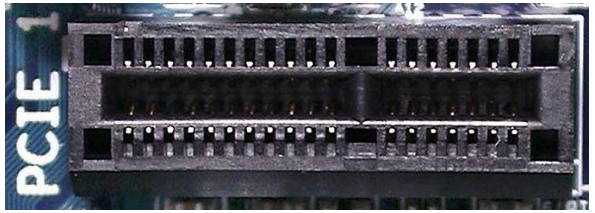
PCIe- und PCI-Slots im Vergleich, by Smial, 2006, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:PCI_und_PCIe_Slots.jpg). CC BY-SA 2.0 Germany.

PCIe Lanes

Hur många ’vägar’ till insticksskortet

Fler **lanes**
→ högre bandbredd

Lanes	Length
x1	25 mm
x4	39 mm
x8	56 mm
x16	89 mm

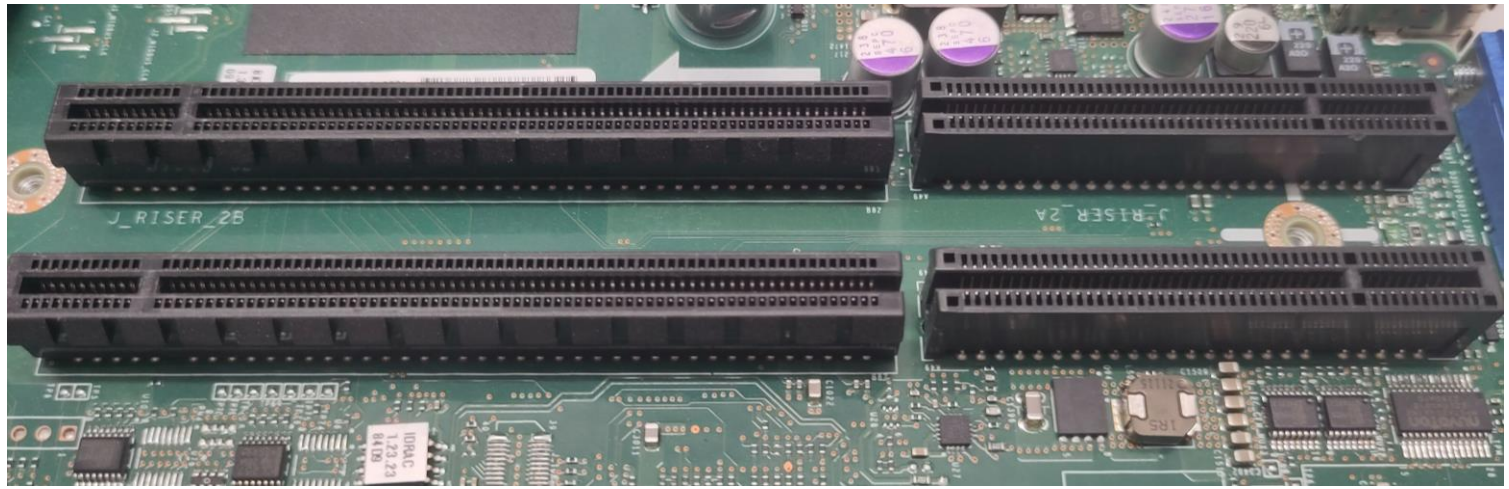


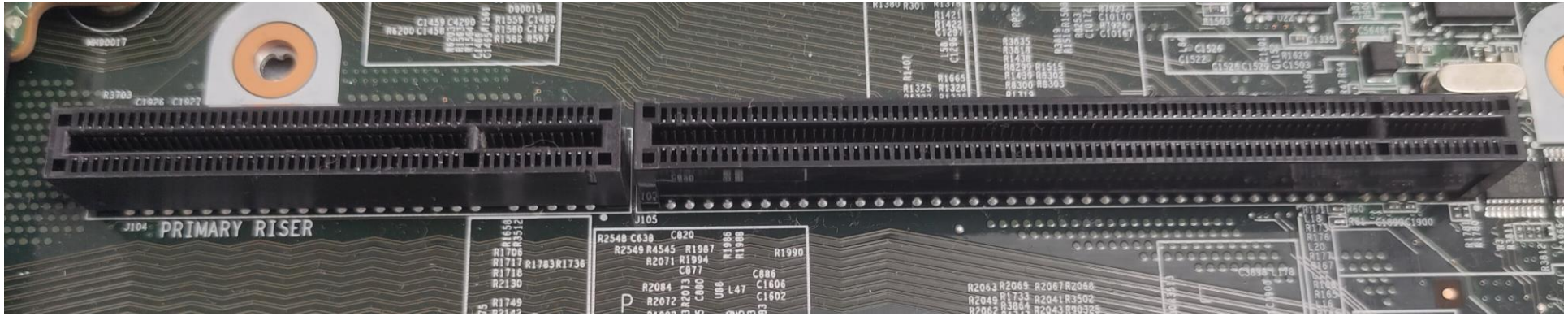
The PCI-Express bus connector with 1 lane, by Clemens PFEIFFER, Vienna, 2005, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:PCI-Express-Bus-1-lane.jpg). CC BY-SA 2.5.

The PCI-Express Bus on a Gigabyte-brand Mainboard picture, by Clemens PFEIFFER, Vienna, 2005, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:PCI-Express-Bus.jpg). CC BY-SA 2.5.

PCIe Lanes

Servrar har ännu fler **lanes**, t.ex. **x24**





PCIe Version

Version	Bandwidth/lane	Release Year
1.0	2 Gb/s	2003
2.0	4 Gb/s	2007
3.0	8 Gb/s	2010
4.0	16 Gb/s	2017
5.0	32 Gb/s	2019

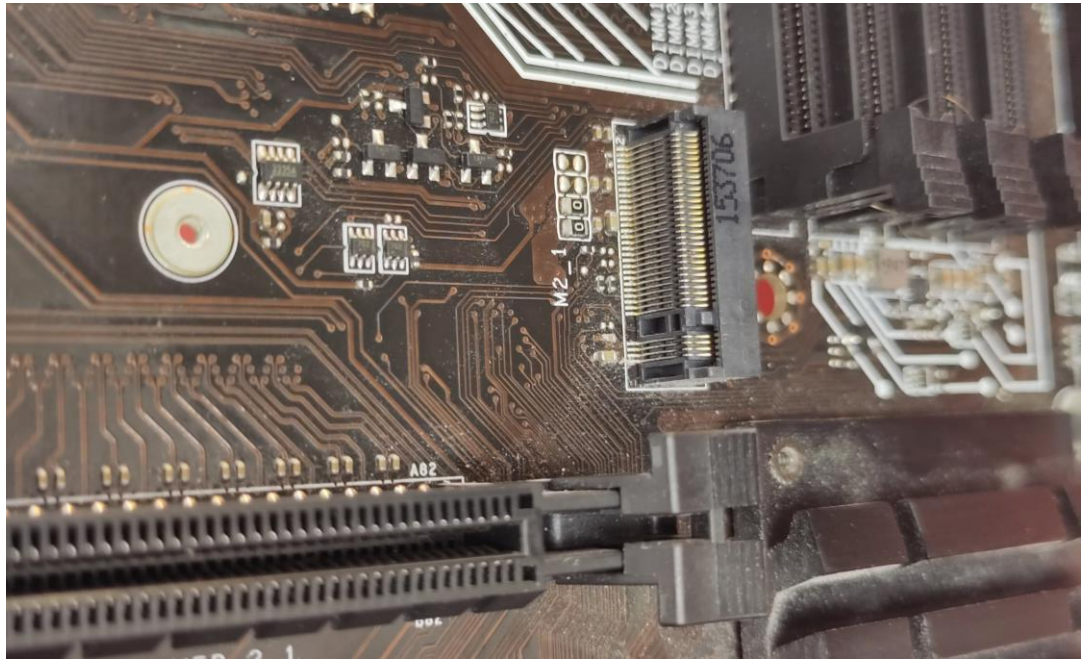
Ex.

PCIe 3.0, x4 lanes

$8 \cdot 4 = 32\, Gb/s$

M.2

A Samsung 980 Pro m.2 PCIe NVMe SSD with 1TB of storage capacity, by D-Kuru, 2022, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:Samsung_980_PRO_PCIe_4.0_NVMe_SSD_1TB-top_PNr%C2%B00915.jpg). CC BY-SA 4.0.



6. Protokoll

SCSI Protocol

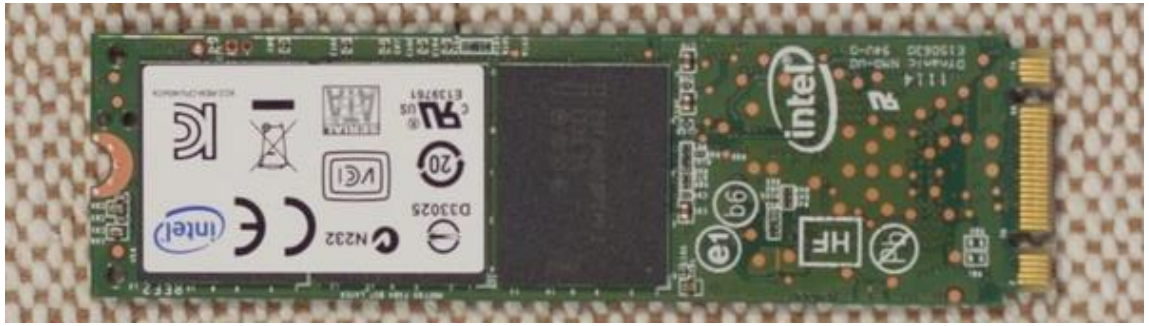
Används ofta i servrar

Modern användning:

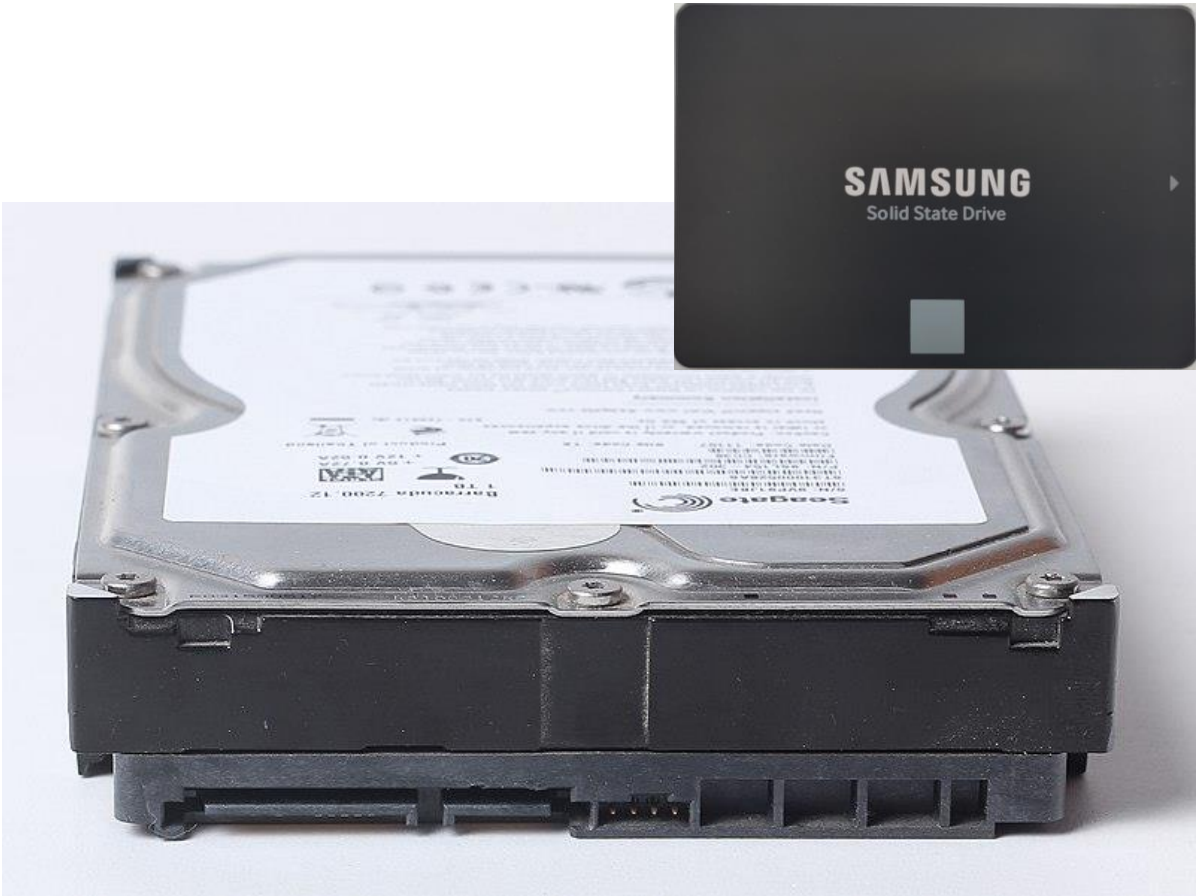
- **SAS**
  Seriell SCSI
- **iSCSI**
  SCSI över IP

Fortfarande bättre prestanda än **SATA**

SATA Protocol



M.2 SSD variants: M.2 SATA (M.2 NGFF) on the left, M.2 NVMe on the right, by ガラパリ, 2021, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:M2_SATA_M2_NVMe_compare.png). CC BY-SA 4.0.



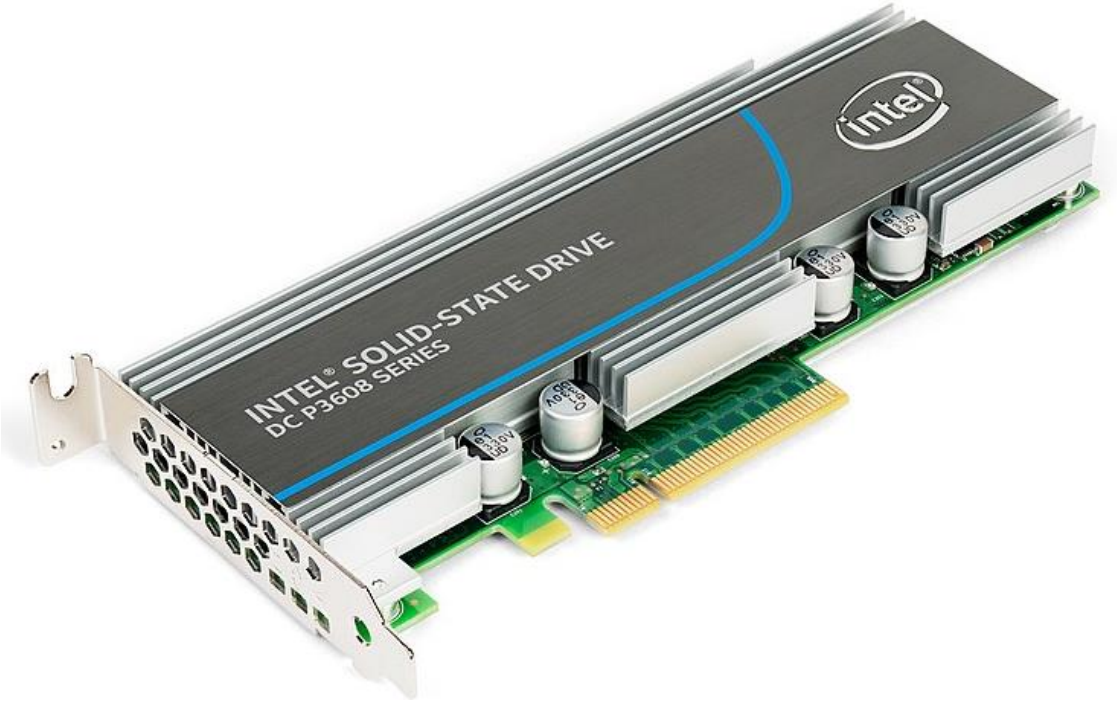
Seagate 3.5" hard disk, by Dmitry Makeev, 2018, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:3.5%22_hard_disk._SATA.jpg). CC BY-SA 4.0.

NVMe

För **N**on-**V**olatile **M**emory (**SSD**)

NVM Express logo, by NVM Express Inc., 2015, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:NVM_Express_logo.svg). Public domain, trademark of NVM Express Inc.

M.2 SSD variants: M.2 SATA (M.2 NGFF) on the left, M.2 NVMe on the right, by ガラパリ, 2021, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:M2_SATA_M2_NVMe_compare.png). CC BY-SA 4.0.



Intel P3608, NVMe-compatible solid-state drive, PCI Express 3.0 x8 low profile add-in card, 1.6TB model, by Dmitry Nosachev, 2016, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:Intel_P3608_NVMe_flash_SSD,_PCI-E_add-in_card.jpg). CC BY-SA 4.0.

7. Chipset & Bandwidth

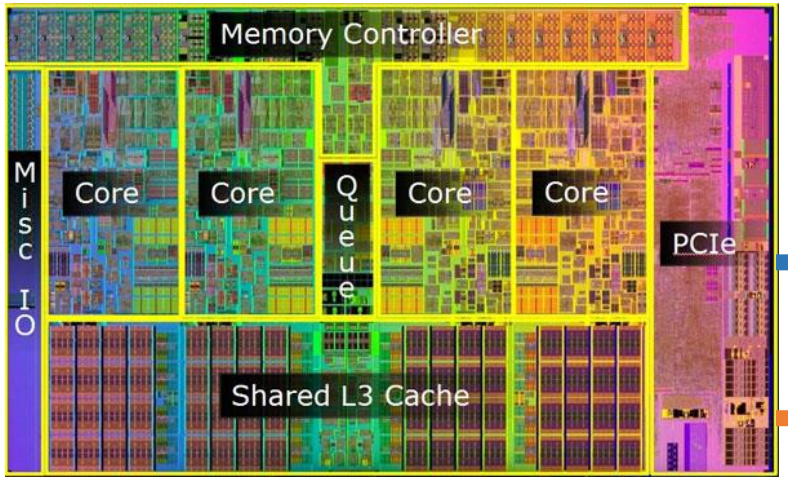
Modern Chipset

Hanterar externa funktioner åt **processorn**

Idag primärt som

- **Multiplexer** för **PCIe**
- **SATA** controller

Modern Chipset



[2]

x16

x16

x16

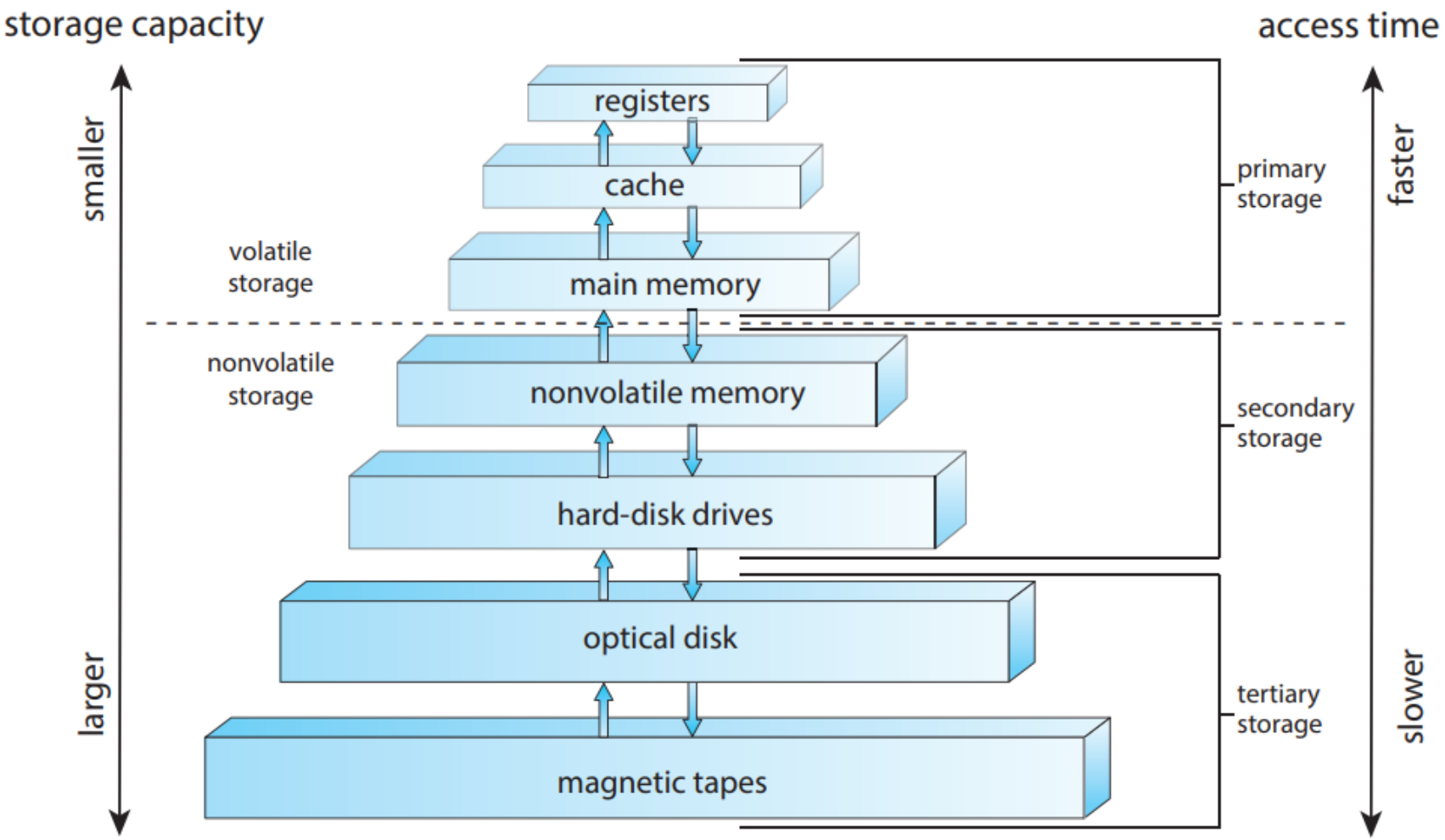
x4

x16

[3]

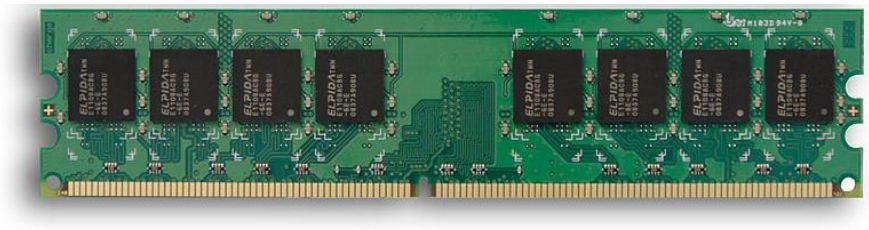
8. Cache

Memory Hierarchi

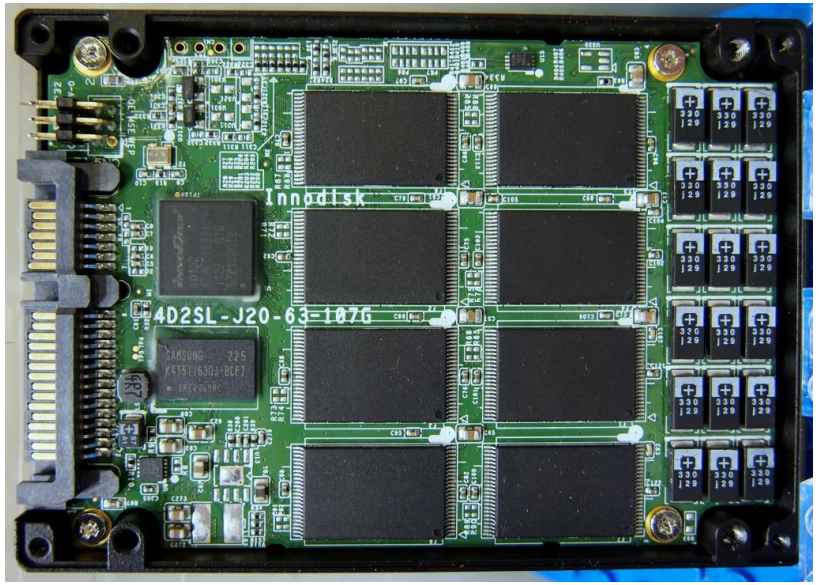


[1]

The top side of an AMD Phenom II X4 840 (HDX840WFK42GM) CPU, by D-Kuru/Wikimedia Commons, 2012, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:AMD_Phenom_II_X4_840_(HDX840WFK42GM)_CPU-top_oblique_PNr%C2%B00373.jpg). CC BY-SA 3.0 Austria.

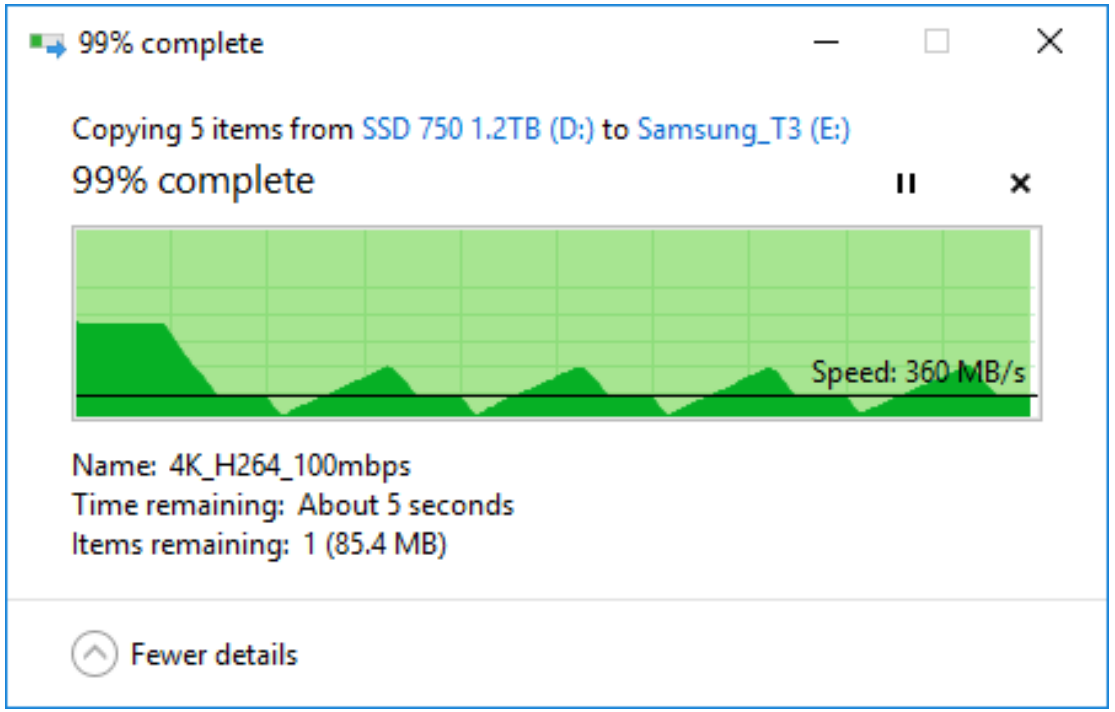


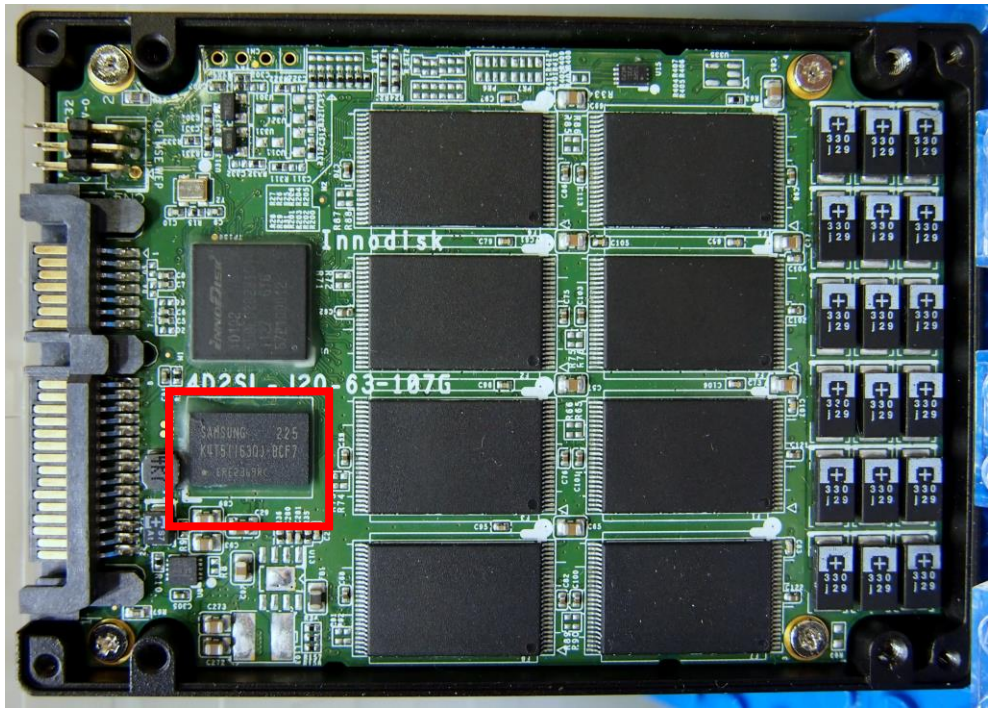
Swissbit 2GB PC2-5300U-555 DDR2 RAM, by An-d, 2013, Wikimedia Commons (https://commons.wikimedia.org/w/index.php?curid=27224495). CC BY-SA 3.0.



Embedded World 2014: Innenleben eines Innodisk-Industrial SSD-Laufwerk, by Ordercracy (Thomas Springer), 2014, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:Embedded_World_2014_SSD.jpg). CC0 1.0.

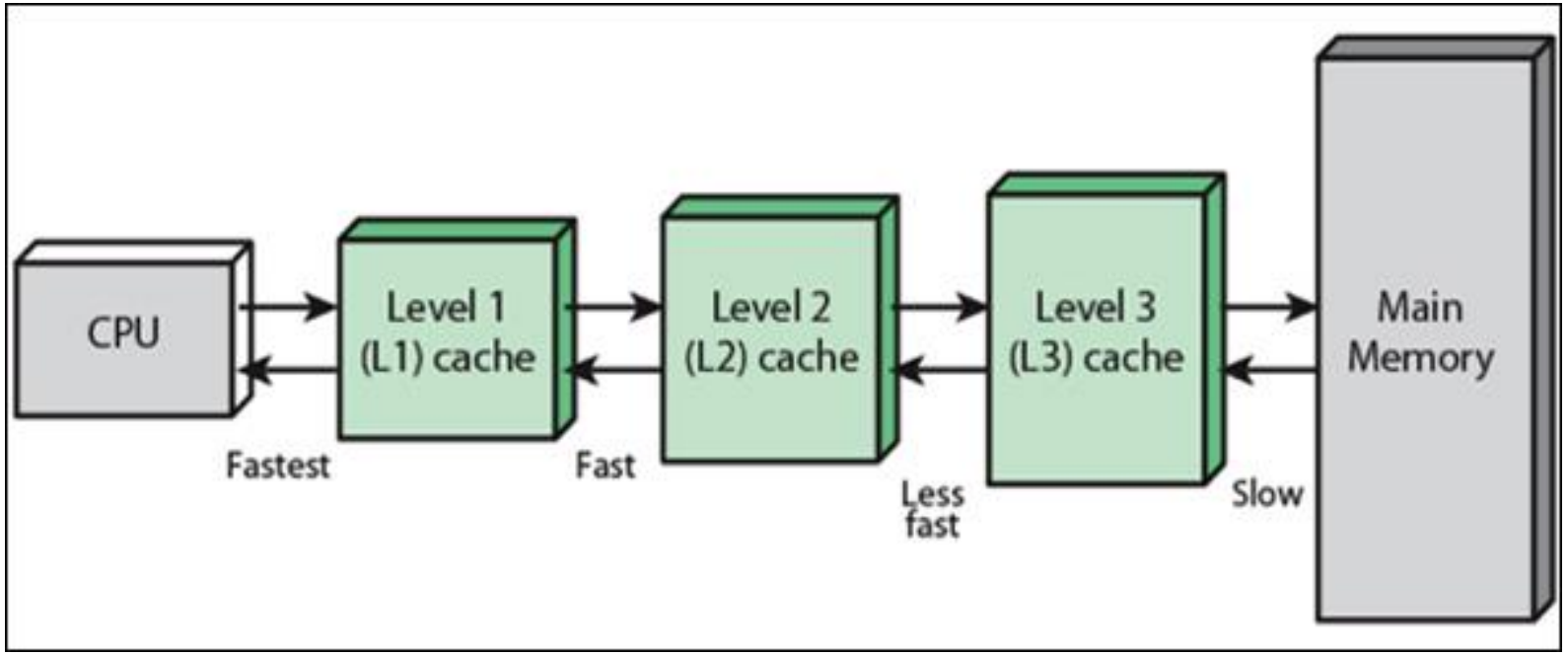
SSD/HDD Cache



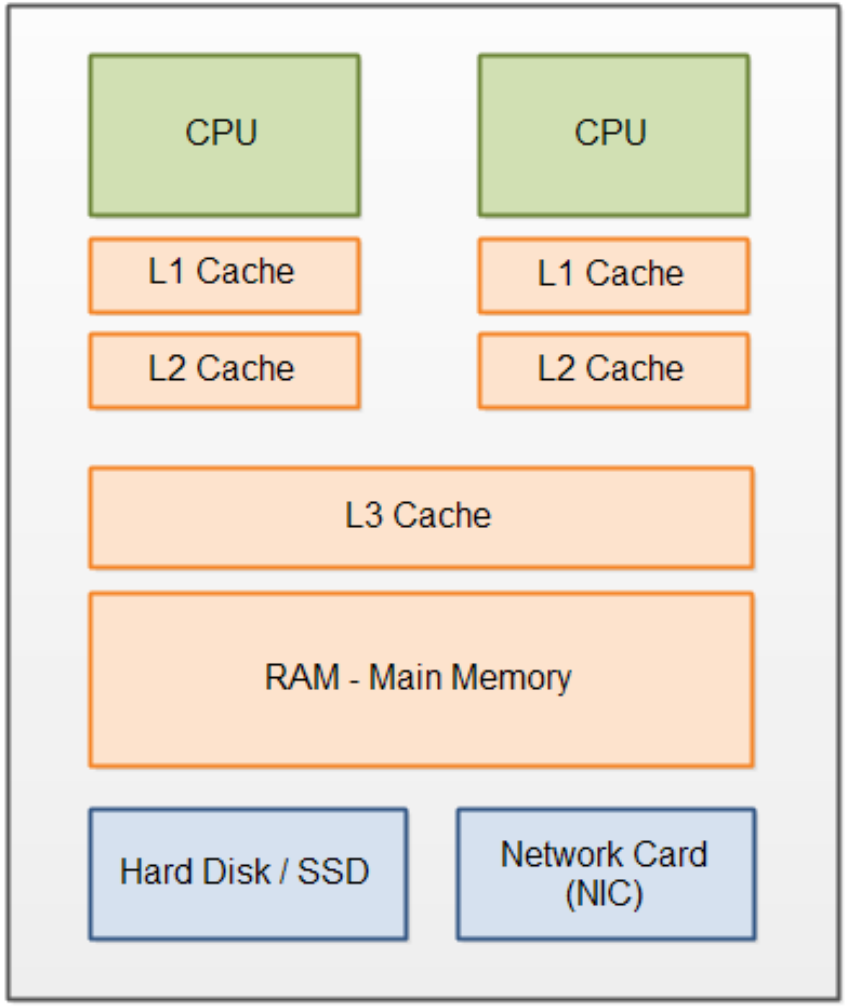


Embedded World 2014: Innenleben eines Innodisk-Industrial SSD-Laufwerk, by Ordercracy (Thomas Springer), 2014, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:Embedded_World_2014_SSD.jpg). CC0 1.0.

CPU Cache


CPU
Level 1
(L1) cache
Level 2
(L2) cache
Level 3
(L3) cache
Main
Memory
Fastest
Fast
Less
fast
Slow

[4]


CPU
CPU
L1 Cache
L1 Cache
L2 Cache
L2 Cache
L3 Cache
RAM - Main Memory
Hard Disk / SSD
Network Card
(NIC)

[5]

Till nästa gång

- Läs kurslitteraturen
- Gör klart laborationen

Tack för idag

Bildreferenser

[1] A. Silberschatz, P. B. Galvin, and G. Gagne, Operating system concepts, 10. ed. Hoboken, NJ: Wiley, 2018.

[2] “LFDDieLayout” [Online image]. Anandtech. Available: https://images.anandtech.com/reviews/cpu/intel/lynnfield/review/LFDDieLayout.jpg. [Accessed: January 8, 2021]

[3] “Asus Zenbook UX32V - motherboard - Intel BD82HM76 (SLJ8E) - Mobile Intel HM76 Express Chipset. Package Size: 25mm x 25mm” [Online image]. Wikimedia Commons. Available: https://commons.wikimedia.org/wiki/File:Asus_Zenbook_UX32V_-_motherboard_-_Intel_BD82HM76_(SLJ8E)-9634.jpg [Accessed: September 28, 2023]

[4] “Cache-memory” [Online image]. Includehelp. Available: https://www.includehelp.com/operating-systems/images/cache-memory.jpg. [Accessed: January 8, 2021]

[5] R. Prabhakar. Cache [Online image]. Softwarerajivprab. Available: https://softwarerajivprab.files.wordpress.com/2019/07/cache.png. [Accessed: January 8, 2021]